

分析化学

Analytical Chemistry

甘峰

“如果觉得有用，请我喝杯茶吧。
咖啡也行。”
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Preface

Before retirement, I wrote two analytical chemistry textbooks: *Fundamentals of Analytical Chemistry* (ISBN: 978-7-5025-9676-7) and *Analytical Chemistry* (ISBN: 978-7-03-063519-8). When writing the former, I lacked experience; when writing the latter, I lacked time. The first was a case of not knowing how to write, and the second a case of not having time to write. Taken together, the two barely qualify as having been written.

Yet a traditional printed textbook is frozen the moment it goes to press. Even when the author discovers errors or hits upon a better way of explaining something, he can only wait for the next revision, and that cycle often stretches over years or even more than a decade. Writing a book is, in the end, an exercise in regret.

After retirement, my time is largely my own, and I have naturally lost my best excuse, that of being too busy. So an idea took hold: why not write, at a comfortable pace, a textbook that truly satisfies me? That is how this book came about.

This book is released in electronic form precisely to break free from the limitations of print. It can be revised, supplemented, and refined at any time, so the reader always receives the latest version. In this sense, it is not a finished book but a book that keeps growing; or rather, a book that can never be fully finished.

Writing a book is no easy task; it weighs on the mind and consumes no small amount of coffee. If you find this book helpful, especially if it spares you from failing a course, consider buying me a coffee ! Your support is what keeps me going.

This book is not without its shortcomings. For any errors or infelicities in the text, I would sincerely appreciate your corrections. My usual email addresses are:

cesgf@mail.sysu.edu.cn and ganfeng1964@163.com. I usually reply as promptly as I can; if I do not, I am most likely off somewhere pretending to survey historic sites, as the cover suggests.

GAN, Feng

Guangzhou, 2026 年 10 月 7 日

Chapter 3 Data Processing in Quantitative Analysis

Chapter Navigation

Repeated determinations of the same sample under identical conditions rarely give exactly the same result. The reason lies mainly in the random error that is unavoidable in any measurement, not in carelessness on the part of the analyst. This chapter follows the complete path of experimental data: Section 1 treats the rules for reading measuring instruments and the principal sources of error, so that data are recorded properly from the outset; Section 2 describes the statistical distribution of random error in large numbers of repeated measurements, which provides the theoretical basis for rejecting data and reporting results; Section 3 then takes up the detection and rejection of outliers and the proper expression of an analytical result, so that a reported value is free of the influence of aberrant data while honestly reflecting the precision of the measurement; Section 4 examines how errors propagate, showing how the error of each step accumulates in the final result; Section 5 covers the correct use of significant figures, so that the recorded data and the computed result match the precision of the measurement; Section 6 introduces statistical tests, the t test and the F test among them, for deciding whether a mean differs significantly from a true value, or whether two or more sets of results differ from one another; and Section 7 discusses the construction of a calibration curve and its regression analysis, by which an instrumental response is converted reliably into the content of the analyte.

The goal of a quantitative analysis is the content of the analyte. When the

analyte is present, that content has a definite value, conventionally called the true value and denoted by the Greek letter μ. The true value, however, cannot be measured directly; we can only measure such physical quantities as mass, volume, and absorbance, and from them calculate the content. This calculation passes through several measurement steps, and every measurement is limited by the accuracy of the instrument and by human factors in the procedure. Error is therefore unavoidable at every step, from the measurement itself to the final conclusion.

A reliable analytical result requires satisfactory answers to four questions in turn.

1. **How is the measurement plan established?** This covers the choice of analytical method, the selection of instruments and equipment, the design of the procedure, and so on. If this step is flawed, no amount of subsequent data processing can repair it.
2. **How are the data treated?** Given raw measured values that contain error, how can the information closest to the true value be extracted? Does error follow any regular pattern? Can a mathematical theory be built to guide the analysis of data?
3. **How is the result expressed?** A complete analytical result is not a bare number; it must be accompanied by a measurement uncertainty that states quantitatively how trustworthy that number is.
4. **How is the result evaluated?** On one hand, the precision and the uncertainty of the result must be assessed; on the other, the result can be verified against a standard method or a certified reference material, and a judgment reached as to whether it meets the intended analytical purpose.

The chapter follows this chain: from carrying out the measurement, through

treating the data, to expressing the result and evaluating its quality. One question runs through it all, namely how far the result can be trusted.

3.1 Sources and Classification of Error

3.1.1 Obtaining the Measured Data

Obtaining measured data looks simple, yet it is one of the most common sources of error. The basic measuring instruments of quantitative analysis, such as the balance, graduated cylinder, volumetric flask, pipet, and burette, all require us to read a scale. According to the way an instrument is graduated, readings fall into two classes: **complete reading** and **estimated reading**.

Consider complete reading first. Figure 3.1 shows a single-pan electronic balance; weighing a sample on it follows a typical sequence:

1. Switch on the balance and confirm that it displays 0.0000 g (if it does not, press the tare key);
2. Place a dry small beaker on the pan, press the tare key, and the reading returns to zero;
3. Add the sample, close the glass doors, and when the reading is essentially stable it shows 1.0610 g.

The last digit, however, fluctuates among 1.0609, 1.0610, and 1.0611. Two questions follow.

1. Should we record 1.061 g or 1.0610 g?
2. While the reading fluctuates, which value should we take?

To the first question: record the full value, 1.0610 g. The final 0 is not su-

perfluous; it states that the balance resolves 0.0001 g. Dropping one digit lowers the precision by an order of magnitude. Many students go wrong here precisely because they overlook the significance of the last digit.

The second question concerns the fluctuation itself. The last digit of a balance drifts now high, now low, in no fixed direction; this is the random fluctuation produced by countless small influences acting throughout the measuring system, not a one-way drift. In macro analysis a fluctuation of ±0.0001 g is entirely acceptable, and any of the three values may be taken, though the central value of the fluctuation is the more reasonable choice.

Now consider estimated reading. Figure 3.2 shows a 50 mL graduated cylinder in which the meniscus lies between the 44 mL and 45 mL marks, with no finer graduation between them. Some would record 44 mL or 45 mL outright. That is not correct. The meniscus has passed 44 mL but has not reached 45 mL; the excess can only be estimated, and a reasonable estimate lies between 44.4 and 44.6 mL. A graduated cylinder is a comparatively coarse instrument, normally used for measuring liquids roughly, and it does not call for fine interpolation. An estimate of 44.1 mL, however, would clearly misplace the meniscus.

Either way, the recorded digits contain an estimated component. In complete reading that component is small, only the last digit varying by ±1; in estimated reading it is larger, the last digit resting on personal judgment. All the meaningful information these digits carry is called the significant figure: for complete reading it is every digit the instrument displays, as in 1.0610 g; for estimated reading it is the certain digits plus one estimated digit, as in 44.5 mL. The essential point is that although the last digit is an estimate, it must be retained; discard it and the information that measures the quality of the measurement is lost. Section 3.5 treats this in detail.

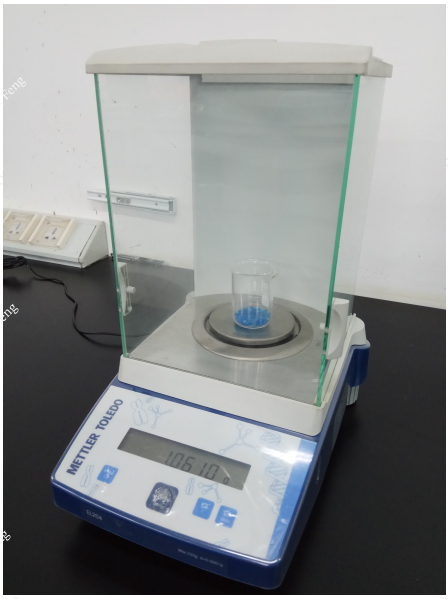


Figure 3.1: Reading a single-pan balance.



Figure 3.2: Reading a graduated cylinder.

Stop and Think

You read 21.37 mL from a 25 mL burette. Is this a complete reading or an estimated reading? Where did the final 7 come from? If a classmate reads 21.36 mL, who is right?

3.1.2 Systematic Error

Systematic error is the component of the deviation that arises from definite causes and appears consistently; it is denoted by ξ. It displaces the measured value from the true value in the same direction each time, as when a weight of incorrect mass makes every weighing deviate the same way.

Systematic error has three characteristics.

- Reproducibility: under the same measurement conditions the error recurs in the same manner;
- Unidirectionality: its sign is constant, either always positive or always negative;
- Detectability: its magnitude and sign can be estimated by comparison with a known standard, and the result can be corrected accordingly.

By the way it varies, systematic error falls into two further classes.

- Constant systematic error: magnitude and sign do not change from measurement to measurement, as with a weight of incorrect mass.
- Variable systematic error: magnitude changes with the measurement conditions in some regular way, as when the graduation of a burette is uneven and the deviation grows in proportion to the volume delivered.

Systematic error in quantitative analysis has four principal sources.

- Method error: an imperfection of the analytical method itself, such as incomplete titration reaction, a color change of the indicator that does not coincide with the equivalence point, or loss of precipitate by dissolution; such effects usually bias the result high or low.
- Instrumental and reagent error: an incorrect zero of an instrument, an error in the graduation of a volumetric vessel, wear of weights, insufficient purity of a reagent, and the like.

- Operational error: error introduced by departure from the specified procedure, such as an unrepresentative sample (a matter of sampling design), an incorrect ignition temperature, or weighing a precipitate before it has cooled.
- Personal error: error arising from the habits of the analyst, such as a preference in judging the color of an indicator, or unconsciously making each titration resemble the one before.

In a given measurement the various systematic errors usually act together. Because they generally do not change with the measurement conditions, they can be removed by correction. The usual means of correction are:

- Instrument calibration: calibrate the balance, pipet, volumetric flask, and burette at regular intervals, and use the corrected values in the calculation;
- Blank determination: replace the sample with distilled water and carry out every step of the procedure; subtract the blank from the result for the sample;
- Comparison:
  - against a certified reference material of similar matrix and known content, comparing the measured value with the certified value;
  - against a standard method, applying the method under test and the accepted standard method to the same sample.

Common Pitfall

Systematic error and random error frequently coexist. Averaging repeated measurements cannot remove systematic error: once present, it recurs in every measurement. Only calibration, a blank determination, or comparison against a reference can eliminate it.

3.1.3 Random Error

Return to the weighing example. The fluctuation of the last digit of the balance reading arises from many small influences in the instrument and its surroundings, such as momentary changes in temperature, humidity, and barometric pressure, fluctuations of current and voltage in the circuit, drafts in the balance room, and slight vibrations of the bench. None of these can be controlled in advance, each varies at random, and all of them affect the output signal of the balance in a random way. A deviation caused by such uncontrollable random factors is called random error and is denoted by ε. It makes a single measured value scatter about the true value with a sign that may be positive or negative and cannot be foreseen.

Random error shows two basic characteristics.

1. Uncertainty: in a single measurement neither the magnitude nor the sign of the random error can be predicted. The n measurements each produce an ε_i ($i = 1, 2, \cdots, n$) whose magnitude and sign are not fixed and which are uncorrelated with one another. It follows that random error cannot be removed from a single result by calibration.
2. Statistical regularity: although the random error of a single mea-

surement cannot be predicted, the results of a large number of repeated measurements follow a definite statistical distribution.

A single random error can neither be predicted nor removed by calibration. Yet the statistical regularity that emerges from many repeated measurements allows us to draw reliable information from such data and to assess its trustworthiness quantitatively, using probability theory and mathematical statistics.

3.1.4 Total Error

In an actual measurement systematic and random error are present together, and a single measured value can be written:

$$x = \mu + \xi + \varepsilon \tag{3.1}$$

Here $\xi + \varepsilon$ is called the total error and is denoted by δ:

$$\delta = \xi + \varepsilon$$

The total error is also called the absolute error. It measures how far the measured value lies from the true value; its sign gives the direction of the deviation and its magnitude gives the size. Dividing the total error by the true value gives the relative error, denoted by δ_r:

$$\delta_r = \frac{\delta}{\mu} \times 100\% \tag{3.2}$$

The relative error relates the error to the magnitude of the quantity measured, and so makes the accuracy of measurements of different size, or made by different methods, comparable.

Example 3.1 Two NaCl samples contain 0.15 g and 0.10 g of the salt. Two analysts determine them and obtain 0.14 g and 0.09 g, respectively. Calculate the absolute and the relative error.

Solution: the absolute errors are

$$\delta_1 = 0.14 - 0.15 = -0.01\,\text{g}$$
$$\delta_2 = 0.09 - 0.10 = -0.01\,\text{g}$$

and the relative errors are

$$\delta_{r_1} = \frac{-0.01}{0.15} \times 100\% = -7\%$$
$$\delta_{r_2} = \frac{-0.01}{0.10} \times 100\% = -10\%$$

The two absolute errors are equal, but the second sample is the smaller one, so its relative error is larger and its accuracy lower. Judging by the absolute error alone would wrongly suggest that the two methods are equally accurate.

Key Concept

The quality of a measurement is judged in two ways. **Accuracy** is the closeness of the result to the true value and is governed by systematic error: the smaller the systematic error, the more accurate the result. **Precision** is the closeness of repeated determinations to one another and is governed by random error: the smaller the random error, the more precise the result. An ideal measurement reduces both.

3.2 Statistical Distribution of Measured Data

Recognizing that random error exists is only the first step in data processing. The more important question is whether random error obeys any law, and, if it

does, what mathematical form that law takes. This section begins with frequency statistics and then sets out the statistical regularity that random error follows.

3.2.1 Frequency Distribution

Everyday experience shows that a repeatedly observed event is more trustworthy, and measurement obeys the same rule. Table 3.1 lists the results of 100 determinations of the iron content of an alloy sample. However, inspecting this set of data by eye alone, one can hardly draw a definite conclusion.

Table 3.1: Iron content of an alloy (mg/L)

1.27	1.30	1.31	1.32	1.32	1.34	1.34	1.34	1.34	1.35
1.35	1.35	1.36	1.36	1.36	1.36	1.36	1.36	1.37	1.37
1.37	1.37	1.37	1.37	1.37	1.37	1.38	1.38	1.38	1.39
1.39	1.39	1.39	1.39	1.39	1.39	1.39	1.39	1.40	1.40
1.40	1.40	1.40	1.40	1.40	1.41	1.41	1.41	1.41	1.41
1.41	1.41	1.42	1.42	1.42	1.42	1.42	1.42	1.42	1.42
1.42	1.42	1.42	1.42	1.42	1.42	1.42	1.43	1.43	1.43
1.43	1.43	1.44	1.44	1.44	1.44	1.44	1.45	1.45	1.45
1.45	1.45	1.45	1.45	1.46	1.46	1.46	1.46	1.46	1.47
1.47	1.47	1.48	1.48	1.48	1.48	1.49	1.50	1.53	1.55

A simple-minded approach is to count the frequency with which each measured value appears, as Figure 3.3 shows, where the most frequent value is 1.42 (also called the mode). Yet to take 1.42 as the best estimate of the true value on this basis has obvious limits: it accounts for only about 15% of the total, so its trustworthiness is slight. More to the point, a measured value can in theory take infinitely dense values, and as the number of measurements tends to infinity the frequency of any single numerical value becomes smaller and smaller and finally

tends to zero. Estimating the true value directly from the frequency of an individual measured value therefore does not hold up.

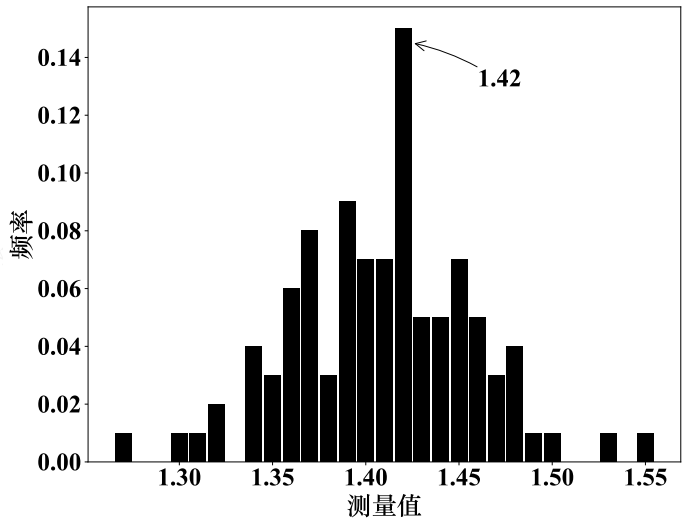


Figure 3.3: Frequency statistics of the measured values.

For this reason statistics no longer attends to the frequency of a single value but counts how many measured values fall in each interval, that is, the frequency distribution. It is constructed as follows:

1. Calculate the range $R = \text{maximum} - \text{minimum}$;
2. Fix the number of class intervals k (usually $k = 10$), with class width $h = R/k$;
3. Assign each interval the grouping rule $L < x \leq U$ (the first interval includes its lower limit);
4. Count the number of measured values in each interval to obtain the frequency distribution table (Table 3.2).

Plotting the frequencies of the intervals in Table 3.2 as a histogram gives Figure 3.4. Table and figure together show clearly that the data display a marked **central tendency** and **dispersion**: most values gather in the middle region and

Table 3.2: Frequency distribution of the iron content of the alloy

Class interval	Frequency (f)	Class midpoint (x_M)	$f \times x_M$
1.270 ~ 1.298	1	1.284	1.28
1.298 ~ 1.326	4	1.312	5.25
1.326 ~ 1.354	7	1.340	9.38
1.354 ~ 1.382	17	1.368	23.26
1.382 ~ 1.410	23	1.396	32.11
1.410 ~ 1.438	20	1.424	28.48
1.438 ~ 1.466	17	1.452	24.68
1.466 ~ 1.494	8	1.480	11.84
1.494 ~ 1.522	1	1.508	1.51
1.522 ~ 1.550	2	1.536	3.07
Total	100	14.100	140.86

thin out toward either side, forming a bell-shaped outline.

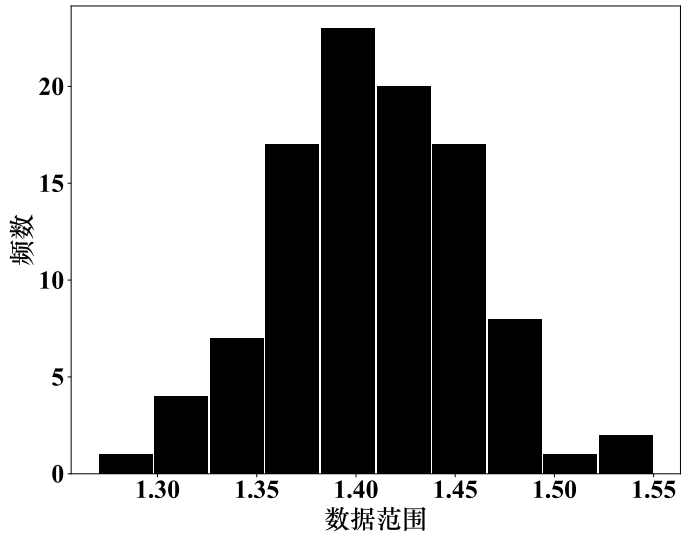


Figure 3.4: Frequency distribution of the iron content of the alloy.

A frequency distribution diagram displays the shape of the data directly, but it raises a deeper question: does this bell shape obey some kind of distribution law that mathematics can describe exactly?

3.2.2 Normal Distribution

To answer that question, consider first how the distribution of measured values relates to the distribution of error. With systematic error already eliminated, a measured value is $x = \mu + \varepsilon$, so its distribution is merely the error distribution translated as a whole to the true value μ, with exactly the same shape. The bell-shaped outline of the measured values in Figure 3.4 therefore portrays the shape of the error distribution itself. Scientists have found this bell shape to be extraordinarily common in nature; human stature and measurement error both show it to some degree. Because probability theory regards it as the canonical form of an error distribution, it was given the name “normal,” and the concept of the

normal distribution followed. When the German mathematician Carl Friedrich Gauss studied errors of astronomical observation, he derived this distribution rigorously from the standpoint of the arithmetic mean, so the normal distribution is often called the Gaussian distribution. Random error accumulates from numerous small and approximately independent contributions, such as temperature, humidity, drafts, and voltage; by the central limit theorem its distribution tends to the normal distribution.

The passage from a frequency distribution to the normal distribution is a passage from the discrete to the continuous. Recall the frequency histogram of Figure 3.4: the height of each rectangle reflects the frequency with which measured values occur in that interval. Divide the frequency by the total number of measurements and by the interval width, and the result is a frequency density; the area of each rectangle then represents a relative frequency. As the number of measurements tends to infinity and the interval width shrinks to zero, the outline of these rectangles smooths into a continuous curve. Mathematically this curve is called a **probability density curve**, and its functional form is the **probability density function**. Its height at any point x measures the relative likelihood that a measured value occurs near x, while the area under the curve between two values of x represents the probability that a measured value falls in that interval.

The precise definition of the probability density function is this: it is a nonnegative function $f(x)$ such that the probability that a measured value falls in any interval $[a, b]$ equals the integral of the function over $[a, b]$, that is,

$$P(a \le X \le b) = \int_a^b f(x)\,\mathrm{d}x$$

The concept is analogous to material density in physics. The density of a piece of metal at each point ($\mathrm{g/cm^3}$) is not itself a mass, but multiplying it by a small volume element and integrating over a region gives the mass of that region;

in the same way probability density is not itself a probability, but integrating it gives a probability.

The probability density function of the normal distribution is

$$f(x) = \frac{1}{\sqrt{2\pi}\sigma} e^{-\frac{(x-\mu)^2}{2\sigma^2}}, \quad -\infty < x < +\infty \tag{3.3}$$

where μ is the true value, σ^2 is the variance, and σ is the standard deviation[1].

Here we introduce the concept of a random variable. Mathematically, a random variable is a function that maps the outcome of a random experiment to a numerical value. Intuitively, it is the quantity subject to randomness in which we are interested, such as the numerical value obtained by measuring some physical quantity. A capital X denotes the random variable, and a lowercase x denotes one particular value it takes.

The normal distribution is written $X \sim N(\mu, \sigma^2)$, with $-\infty < \mu < \infty$ and $\sigma > 0$. Figure 3.5 shows two features of the normal density curve: (1) μ fixes the position of the center of the curve; (2) σ fixes its width, a larger σ giving a flatter curve and a smaller σ a narrower, taller one.

The total area under the normal density curve is always 1. This is exactly what the definition of a probability density function requires:

$$\int_{-\infty}^{+\infty} f(x)\,\mathrm{d}x = 1 \tag{3.4}$$

A larger σ therefore makes the curve lower (see Figure 3.6). This also explains why σ can serve as an index of precision: the larger σ is, the more scattered the data and the poorer the precision of the measurement.

The density function describes the shape of a distribution, but in practical calculation we more often need the probability that X does not exceed some value.

[1]This book follows the Chinese national standard GB/T 3358 and uses “standard deviation” throughout.

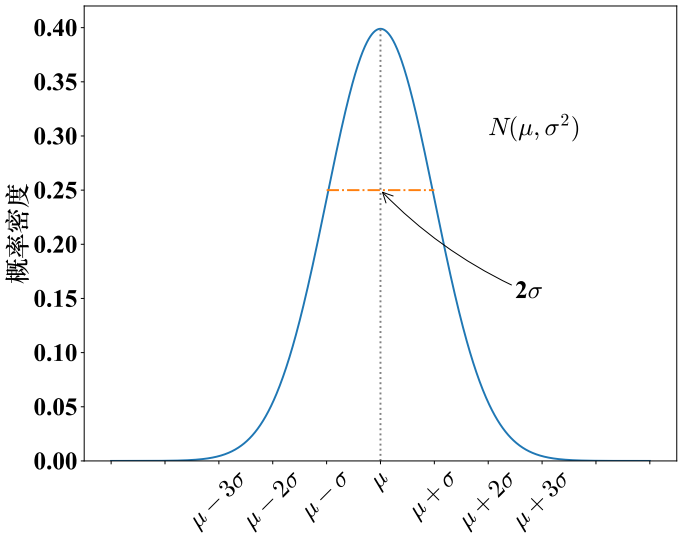


Figure 3.5: Normal distribution.

The function that describes this probability is the cumulative distribution function:

$$F(x) = P(X \leq x) = \frac{1}{\sqrt{2\pi}\sigma} \int_{-\infty}^{x} e^{-\frac{(x-\mu)^2}{2\sigma^2}} \, \mathrm{d}x \tag{3.5}$$

Here $F(x)$ is the area enclosed by the density curve from $-\infty$ to x, that is, the cumulative probability.

For a continuous random variable the probability of a single point is always 0 ($P(X = x) = 0$). Only interval probabilities concern us:

$$P(a < X \leq b) = \frac{1}{\sqrt{2\pi}\sigma} \int_{a}^{b} e^{-\frac{(x-\mu)^2}{2\sigma^2}} \, \mathrm{d}x \tag{3.6}$$

The trouble is that $F(x)$ depends on the two parameters μ and σ, so a different experiment requires the integral to be computed again. For convenience, statistics introduces the **standardizing transformation**, by which any normal distribution $N(\mu, \sigma^2)$ can be converted into one common distribution. Let

$$U = \frac{X - \mu}{\sigma} \tag{3.7}$$

Then the cumulative distribution function of U is

$$F_U(u) = P(U \le u) = P\left(\frac{X-\mu}{\sigma} \le u\right) = P(X \le \mu + \sigma u) = F_X(\mu + \sigma u) \tag{3.8}$$

and the probability density function $\phi(u)$ of U is

$$\phi(u) = \frac{\mathrm{d}}{\mathrm{d}u} F_U(u) = \frac{\mathrm{d}}{\mathrm{d}u} F_X(\mu + \sigma u) = f_X(\mu + \sigma u) \times \sigma = \frac{1}{\sqrt{2\pi}} e^{-\frac{u^2}{2}} \tag{3.9}$$

In other words, U follows the $N(0,1)$ distribution, also called the **standard normal distribution** (the solid curve in Figure 3.6). Once the standard normal distribution is available, a general probability table can be prepared (Table 3.3), and any problem involving a normal distribution can be solved by standardizing and looking up the table.

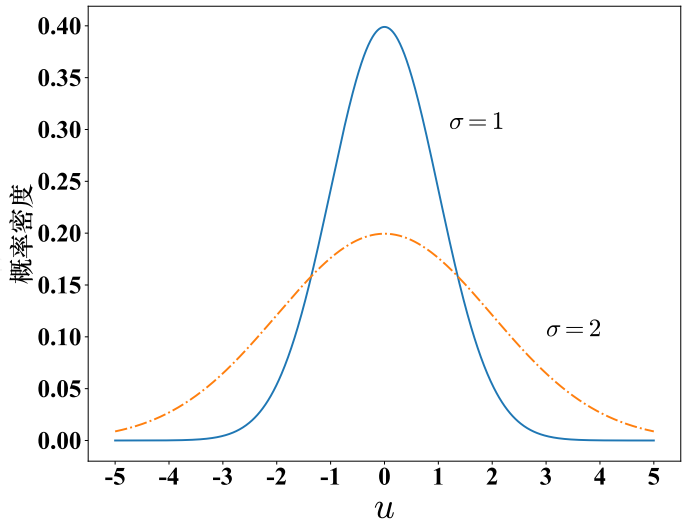


Figure 3.6: Standard normal distribution.

Table 3.4 gives the interval probability for three values of u. The key conclusion is that a measured value falls within $\mu \pm 2\sigma$ with a probability of 95.5%. This shows that setting an interval with σ as the basis and estimating the true value from it is workable; its rigorous statistical form is the confidence interval treated later in this chapter.

Table 3.3: Probability table for the standard normal distribution

$\|u\|$	Area	$\|u\|$	Area	$\|u\|$	Area
0.0	0.0000	1.0	0.3413	2.0	0.4772
0.1	0.0398	1.1	0.3643	2.1	0.4821
0.2	0.0793	1.2	0.3849	2.2	0.4861
0.3	0.1179	1.3	0.4032	2.3	0.4893
0.4	0.1554	1.4	0.4192	2.4	0.4918
0.5	0.1915	1.5	0.4332	2.5	0.4938
0.6	0.2257	1.6	0.4452	2.6	0.4953
0.7	0.2580	1.7	0.4554	2.7	0.4965
0.8	0.2881	1.8	0.4641	2.8	0.4974
0.9	0.3159	1.9	0.4713	3.0	0.4987

This “95% certainty” has a formal name in statistics, the **confidence level** (denoted P), which states how firmly we assert that a given interval covers the true value. The concept complementary to it is the **significance level** (denoted α), for which $P = 1 - \alpha$. Taking $P = 95\%$ means $\alpha = 5\%$: we are 95% confident that the conclusion is correct, and we accept a 5% risk of misjudgment. Grasping this pair of concepts is the common foundation of the outlier tests, the t test, and interval estimation that follow.

Table 3.4: Interval probability of measured values

$\|u\|$	Interval of the measured value	Interval probability
1.0	$x = \mu \pm 1\sigma$	68.3%
2.0	$x = \mu \pm 2\sigma$	95.5%
3.0	$x = \mu \pm 3\sigma$	99.7%

A frequency distribution diagram is no more than an intuitive description of

the data, and the probabilities it gives are discrete and approximate; the normal distribution, by contrast, provides a continuous theoretical model that makes the probability of a measured value falling in any interval exactly computable. Before the model is used, however, the meaning of its two parameters μ and σ has to be clear.

3.2.3 Meaning of the Parameters of the Normal Distribution

The normal distribution is completely controlled by the two parameters μ and σ^2. What is their statistical meaning? The question calls for the concept of mathematical expectation. For a continuous random variable X, the mathematical expectation is defined as

$$E[X] = \int_{-\infty}^{+\infty} x f(x)\,\mathrm{d}x \tag{3.10}$$

For the standard normal variable $U = (X - \mu)/\sigma$, whose probability density function is given in Eq. (3.9), we therefore have

$$E[U] = \int_{-\infty}^{+\infty} u\phi(u)\,du = \int_{-\infty}^{+\infty} u\frac{e^{-\frac{u^2}{2}}}{\sqrt{2\pi}}\,du = 0 \tag{3.11}$$

Since $X = \mu + U\sigma$, the linearity of the mathematical expectation gives

$$E[X] = E[\mu + U\sigma] = E[\mu] + E[U\sigma] = \mu \tag{3.12}$$

Similarly, the variance of X is defined as

$$E\left[(X - E[X])^2\right] = \int_{-\infty}^{+\infty} (x-\mu)^2 f(x)\,\mathrm{d}x \tag{3.13}$$

From the properties of U,

$$E[(U - E(U))^2] = \int_{-\infty}^{+\infty} (u-0)^2 \cdot \frac{1}{\sqrt{2\pi}} e^{-\frac{u^2}{2}}\,du = 1 \tag{3.14}$$

and hence

$$E[(X - E[X])^2] = E[(\mu + U\sigma - \mu)^2] = E[U^2\sigma^2] = \sigma^2 E[U^2] = \sigma^2 \tag{3.15}$$

Equation (3.12) shows that μ is the mathematical expectation of the random variable X. Equation (3.15) shows that σ^2 is the mathematical expectation of the square of the difference between X and its expected value. In measurement the systematic error has been removed by calibration or by a blank determination, and positive and negative random errors are equally likely, so that the random error has zero expectation, $E[\varepsilon] = 0$. From $x = \mu + \varepsilon$ it follows that $E[x] = \mu$: the expectation above is the true value of the analyte, and the parameter μ of the normal distribution thereby acquires a definite physical meaning.

Equations (3.10) and (3.13) give μ and σ^2 in integral form, and their evaluation requires the probability density $f(x)$. Measurement, however, supplies a finite set of observed values $x_1, x_2, \cdots, x_N$, so the integral form must still be converted into a sum.

The weights in the integral are given by the probability density $f(x)$: the product $f(x)\,\mathrm{d}x$ is the probability that a measured value falls within an infinitesimal interval about x. In data, the weights are given by relative frequencies. Divide the real line into a number of small intervals; let c_k be the midpoint of the kth interval and n_k the number of the N measurements falling in it, and define the relative frequency as

$$f_k = \frac{n_k}{N}$$

By the law of large numbers, for N sufficiently large the relative frequency f_k converges in probability to the probability that a measured value falls in that interval, $p_k = P(c_k - h/2 < X \le c_k + h/2)$; that is, the probability that f_k lies near p_k tends to 1 as N increases. Taking f_k as the weight and c_k as the value representing the

interval, the integral of Eq. (3.10) corresponds to

$$E[X] \approx \sum_k c_k f_k = \frac{\sum_k c_k n_k}{N}$$

First let the interval width $h \to 0$, so that the right-hand side tends to the integral $\int x f(x)\,\mathrm{d}x$, that is, to $E[X]$; then let $N \to \infty$, so that by the law of large numbers $\frac{1}{N}\sum_{j=1}^{N} X_j$ converges in probability to $E[X]$. Combining the two gives

$$\mu = \lim_{N\to\infty} \frac{1}{N} \sum_{j=1}^{N} X_j \tag{3.16}$$

The limit here is convergence in probability; textbooks commonly write it as an ordinary limit.

The integral of Eq. (3.13) admits the same discretization. Replacing the summand x by $(x-\mu)^2$ gives the population variance in sum form,

$$\sigma^2 = \lim_{N\to\infty} \frac{1}{N} \sum_{j=1}^{N} (X_j - \mu)^2 \tag{3.17}$$

Both equations presuppose $N \to \infty$. In quantitative analysis the number of measurements is finite; how the population parameters μ and σ are to be estimated from the sample data of a finite number of measurements is the subject of the next section.

3.2.4 Inferring the Population from a Sample

Only a finite number of measurements can be made in practice. Regarding those few measurements as one sampling from an infinite population introduces the concepts of the population and the sample. The population is the complete set of all possible measured values, the sample is the finite body of data actually obtained, and the sample size n is the number of measurements. A sample carries

information about the population, but the information is incomplete. This is the fundamental contradiction that statistical inference must resolve.

Let X follow $N(\mu, \sigma^2)$, and draw from it a sample of size n, $x_1, x_2, \cdots, x_n$. We now define a statistic called the sample mean, as follows:

$$\bar{x} = \frac{\sum_{i=1}^{n} x_i}{n} \tag{3.18}$$

Replacing μ in Eq. (3.17) by $\bar{x}$ gives a preliminary statistic that describes the degree of dispersion:

$$s_n^2 = \frac{\sum_{j=1}^{n} (x_j - \bar{x})^2}{n} \tag{3.19}$$

Here s_n^2 is called the second-order sample central moment in statistics, and s_n is its square root. Note that s_n^2 is *not* an unbiased estimate of the population variance σ^2.

To see the relation between sample and population concretely, let us treat the 100 data in Table 3.1 for the moment as a population. Averaging all the data by Eq. (3.18) gives a population mean of $\mu = 1.41$, and Eq. (3.17) then gives a population standard deviation of $\sigma \approx 0.05$.

Next we draw 10 samples at random, with replacement, from this population, each sample containing 6 measured values, and calculate the mean and the standard deviation of each. The results appear in Table 3.5 .

Averaging the 10 sample means in turn gives $\bar{\bar{x}} = 1.41$, approximately the population mean $\mu = 1.41$. Treating the 10 sample means themselves as one set of data and computing their standard deviation gives $s_{\bar{x}} = 0.01$, which describes the fluctuation of the sample mean from one sampling to the next and is far smaller than the population standard deviation $\sigma \approx 0.05$.

Figure 3.7 shows the distribution of the means of 1000 samples drawn from this population (each still of size 6), with the fitted normal curve drawn as a

Table 3.5: Sampled data and their statistics

Sample	Sampled data						Mean	Standard deviation
1	1.53	1.37	1.41	1.46	1.35	1.34	1.41	0.07
2	1.48	1.41	1.39	1.43	1.42	1.39	1.42	0.03
3	1.42	1.45	1.46	1.44	1.34	1.37	1.41	0.05
4	1.35	1.46	1.41	1.36	1.42	1.37	1.40	0.04
5	1.41	1.42	1.44	1.47	1.37	1.41	1.42	0.03
6	1.45	1.41	1.39	1.45	1.44	1.27	1.40	0.07
7	1.45	1.46	1.43	1.39	1.41	1.38	1.42	0.03
8	1.44	1.41	1.42	1.32	1.45	1.45	1.42	0.05
9	1.34	1.47	1.37	1.48	1.45	1.47	1.43	0.06
10	1.55	1.36	1.34	1.34	1.38	1.49	1.41	0.09

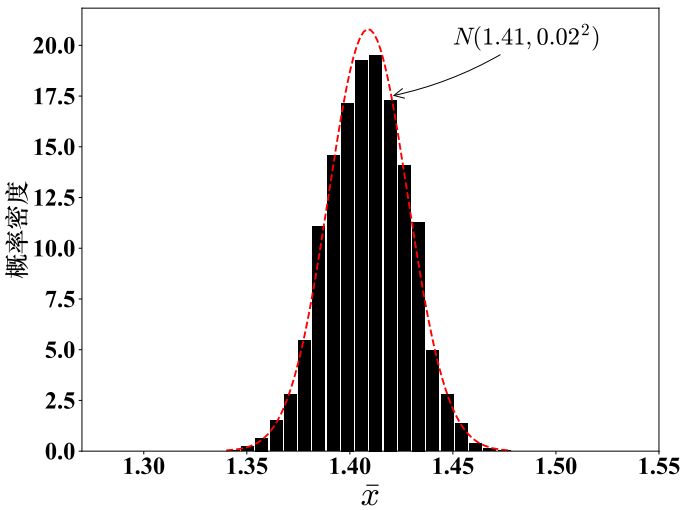


Figure 3.7: Sampling distribution of the sample mean.

dashed line. Compared with the distribution of single measurements, the distribution of the means has a narrower peak, which shows that estimating the true value by a sample mean is more precise than a single measurement.

The result above can be verified theoretically, as follows. Consider first the expectation of the mean:

$$E(\bar{X}) = E\left[\frac{1}{n}\sum_{i=1}^{n} X_i\right] = \frac{1}{n}\sum_{i=1}^{n} E\left[X_i\right] = \frac{1}{n}\sum_{i=1}^{n} \mu = \mu \tag{3.20}$$

The expectation of $\bar{X}$ is thus the population mean μ itself, so $\bar{x}$ is an unbiased estimate of μ. Consider next the variance of the mean:

$$\mathrm{Var}(\bar{X}) = \mathrm{Var}\left(\frac{1}{n}\sum_{i=1}^{n} X_i\right) = \frac{1}{n^2}\sum_{i=1}^{n} \mathrm{Var}(X_i) = \frac{1}{n^2}\sum_{i=1}^{n} \sigma^2 = \frac{\sigma^2}{n} \tag{3.21}$$

where $\mathrm{Var}(X) = E\left[(X - E[X])^2\right]$ is the variance operator, the quantity defined by Equation 3.13 in Section 3.3.

The theoretical result shows that $\bar{x}$ follows $N(\mu, \sigma^2/n)$, with a standard deviation of only $\sigma/\sqrt{n}$. The larger the number of measurements n, the more precise the estimate.

In practice σ is unknown, and s_n is the obvious substitute. Computing the mathematical expectation of s_n^2, however, exposes a problem:

$$\begin{aligned} E[s_n^2] &= E\left[\frac{1}{n}\sum_{i=1}^{n}(X_i - \bar{X})^2\right] \\ &= \frac{1}{n}E\left[\sum_{i=1}^{n}(X_i - \mu)^2 - n(\bar{X} - \mu)^2\right] \\ &= \frac{1}{n}\left(n\sigma^2 - n \cdot \frac{\sigma^2}{n}\right) \\ &= \frac{n-1}{n}\sigma^2 \end{aligned} \tag{3.22}$$

The result is $E[s_n^2] = \frac{n-1}{n}\sigma^2 \neq \sigma^2$: s_n^2 systematically underestimates σ^2 and is a biased estimate. The correction suggests itself, namely multiplying by the factor

$\frac{n}{n-1}$ to give a sample variance with $n-1$ in the denominator:

$$s^2 = \frac{\sum_{i=1}^{n}(x_i - \bar{x})^2}{n-1} \tag{3.23}$$

Then

$$E[s^2] = E\Big[\frac{n}{n-1}\, s_n^2\Big] = \frac{n}{n-1} \cdot \frac{n-1}{n}\,\sigma^2 = \sigma^2$$

where $n-1$ is called the number of degrees of freedom, denoted ν. This proves that s^2 is an unbiased estimate of σ^2.

s^2 is an unbiased estimate of σ^2, and what is usually called the sample variance is precisely this statistic; s is the sample standard deviation, and the standard deviation of the mean is $s/\sqrt{n}$, also called the standard error. In addition, the relative standard deviation s_r is often used to compare the dispersion of data of different magnitude, defined as

$$s_r = \frac{s}{\bar{x}} \times 100\% \tag{3.24}$$

Key Concept

Estimating population parameters rests on just two quantities. One is the use of $\bar{x}$ to estimate μ; the estimate is unbiased and grows more accurate as n increases, with precision improving as $\sqrt{n}$. The other is the use of $s^2 = \sum(x_i - \bar{x})^2/(n-1)$ to estimate σ^2, where the denominator is written $n-1$ precisely so that the estimate is unbiased. Why $n-1$ rather than n? Because $\bar{x}$ is itself computed from these n data, which consumes one degree of freedom: of the n deviations $(x_i - \bar{x})$, only $n-1$ vary independently.

3.2.5 The Three Major Sampling Distributions

The normal distribution handles most problems but does not cover every situation. For different settings statisticians developed three important sampling

distributions, namely the χ^2, t, and F distributions, which together form the theoretical basis of classical statistical inference.

The χ^2 distribution came first. The physicist Maxwell encountered the distribution of a sum of squares of independent normal variables while studying molecular speeds. In 1900 Pearson formally proposed the distribution while working with biological data and established the χ^2 goodness-of-fit test. If $X_1, \cdots, X_n$ are independent and identically distributed as $N(0,1)$, then

$$Q = X_1^2 + \cdots + X_n^2 \sim \chi^2(n), \quad f(q) = \frac{(1/2)^{\frac{n}{2}}}{\Gamma(n/2)} q^{\frac{n}{2}-1} e^{-\frac{q}{2}}, \ q > 0 \tag{3.25}$$

where Γ is the gamma function.

Figure 3.8 shows that the χ^2 distribution is skewed to the right but approaches normality as n increases. The χ^2 distribution yields an important corollary. The sample variance s^2 drawn from a normal population and the population variance σ^2 satisfy

$$\frac{(n-1)s^2}{\sigma^2} \sim \chi^2(n-1) \tag{3.26}$$

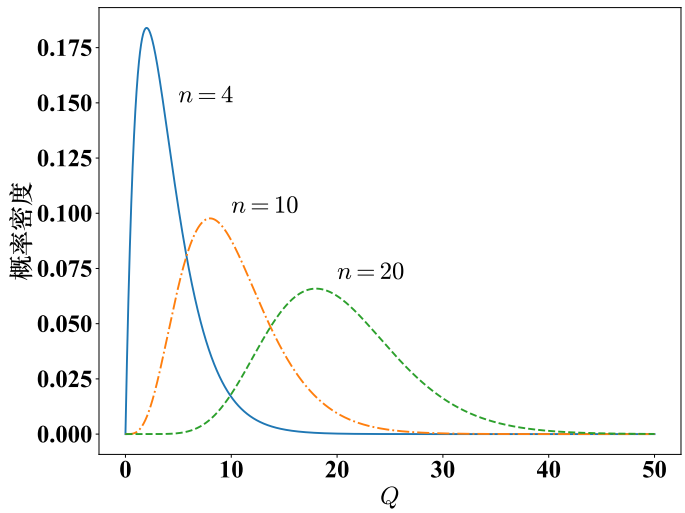


Figure 3.8: The χ^2 distribution.

The appearance of the t distribution is closely bound up with the work of William Sealy Gosset, who was analyzing data at a brewery and usually handled samples of only a few to a dozen measured values. He found that such small samples often did not follow a normal distribution exactly. He later studied statistics under Pearson and eventually established the theory of small-sample analysis.

Suppose $X_1 \sim N(0,1)$ and $X_2 \sim \chi^2(n)$ are independent. Then the statistic

$$t = \frac{X_1}{\sqrt{X_2/n}}$$

follows a t distribution with n degrees of freedom:

$$f(t) = \frac{\Gamma(\frac{n+1}{2})}{\sqrt{n\pi}\Gamma(\frac{n}{2})}\left(1+\frac{t^2}{n}\right)^{-\frac{n+1}{2}}, \quad -\infty < t < +\infty. \tag{3.27}$$

Figure 3.9 compares the t distribution with the standard normal distribution: the t distribution has a lower center and heavier tails. This is exactly how the extra uncertainty of a small sample shows itself. Once $n \geq 30$, however, the t distribution may be replaced approximately by $N(0,1)$.

The t distribution yields one corollary of the first importance. For a sample from $N(\mu, \sigma^2)$,

$$t = \frac{\bar{x}-\mu}{s/\sqrt{n}} \sim t(n-1) \tag{3.28}$$

This is the core formula of the t test, which uses a sample mean to test the significance of the difference between the sample mean and the population mean.

If two independent samples $x_1, x_2, \cdots, x_m$ and $y_1, y_2, \cdots, y_n$ are drawn from the same population, and the sample means of the two sets are denoted by $\bar{x}$ and $\bar{y}$, with sample variances

$$s_x^2 = \frac{1}{m-1}\sum_{i=1}^{m}(x_i-\bar{x})^2 \quad \text{and} \quad s_y^2 = \frac{1}{n-1}\sum_{j=1}^{n}(y_j-\bar{y})^2$$

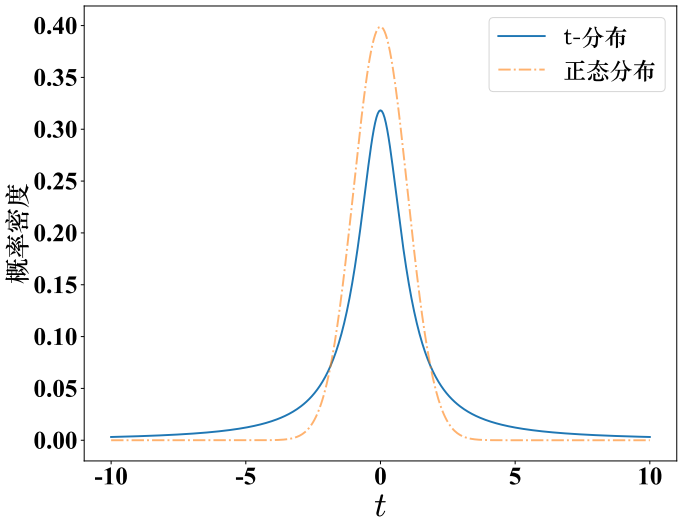


Figure 3.9: Probability density of the t distribution and the normal distribution.

we set

$$s_p^2 = \frac{(m-1)s_x^2 + (n-1)s_y^2}{m+n-2} \tag{3.29}$$

then

$$\frac{(\bar{x}-\bar{y})}{s_p\sqrt{\frac{1}{m}+\frac{1}{n}}} \sim t(m+n-2) \tag{3.30}$$

Equation (3.30) is the core of the t test for two independent samples, used to compare two sets of measured means for a significant difference. Compared with the one-sample test of Eq. (3.28), it makes two adjustments. The first concerns the numerator, where $\bar{x}-\mu$ is replaced by $\bar{x}-\bar{y}$. If the two sets of data come from the same population, $\bar{x}$ and $\bar{y}$ are both unbiased estimates of the population mean μ, and their difference fluctuates at random about 0. The farther the difference departs from 0, the more clearly the premise of a common population fails.

The second adjustment concerns the s_p in the denominator, obtained as the square root of the pooled variance s_p^2, which is a degrees-of-freedom weighted average of the two sample variances. Since the two sets are assumed to come from

the same population, their variances ought to be equal, and pooling the deviation information of both gives a more accurate estimate of that common variance than either variance used alone. The weights are the degrees of freedom $m-1$ and $n-1$: the larger the sample, the more trustworthy its variance estimate and the larger its share in the pooled result. The pooled degrees of freedom are $(m-1)+(n-1) = m+n-2$, which is the degrees of freedom of the t distribution in Eq. (3.30). The meaning is that each sample spends one degree of freedom in computing its mean, so only $m+n-2$ of the $m+n$ deviations vary independently.

This test requires the two samples to be independent of each other and both to come from normal populations of equal variance. When the variances are unequal, pooling loses its basis, so in practice an F test is usually carried out first to judge whether the variances are homogeneous before Eq. (3.30) is applied. The procedure and worked examples appear in Section 3.6.2.

Stop and Think

Two sets of determinations of iron content, one your own ($n = 6$) and one your classmate's ($m = 8$), give different means. Can Eq. (3.30) be applied directly to decide whether they differ? Before calculating, note that besides independence and normality one more premise is missing. If the standard deviation of one set is clearly larger than that of the other, is a pooled variance still reasonable?

Ronald Aylmer Fisher made Gosset's heuristic derivation rigorous and supplied a complete argument. In Gosset's original manuscript, moreover, the statistic was written z; Fisher changed it to t, and the t distribution is known by that name to this day. Fisher's more important contribution was the distribution of the ratio of two variances, which plays a central role in the F test and was later named the F distribution by George W. Snedecor in Fisher's honor. If $X_1 \sim \chi^2(m)$

and $X_2 \sim \chi^2(n)$ are mutually independent, then

$$F = \frac{X_1/m}{X_2/n}$$

follows an F distribution with (m, n) degrees of freedom:

$$f(F) = \frac{\Gamma(\frac{m+n}{2})\left(\frac{m}{n}\right)^{\frac{m}{2}}}{\Gamma(\frac{m}{2})\Gamma(\frac{n}{2})} F^{\frac{m}{2}-1}\left(1+\frac{m}{n}F\right)^{-\frac{m+n}{2}} \tag{3.31}$$

Figure 3.10 shows that the F distribution is skewed to the right but approaches normality as the degrees of freedom increase.

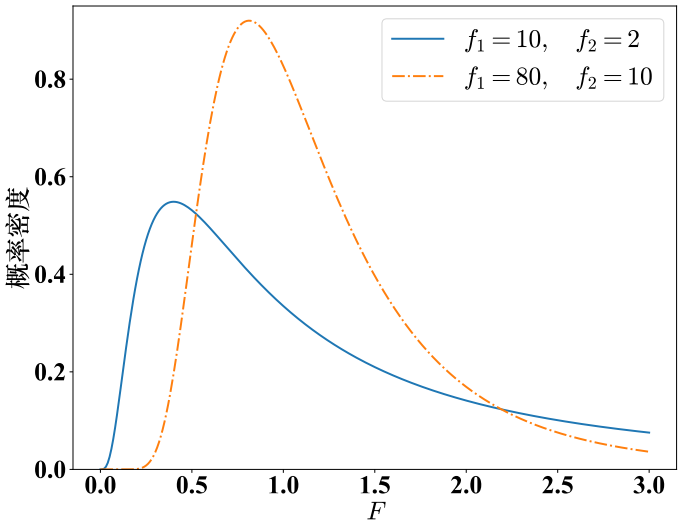


Figure 3.10: Probability density of the F distribution.

The F distribution yields a corollary of greater importance. If independent samples are drawn from two normal populations, then

$$\frac{s_x^2/\sigma_1^2}{s_y^2/\sigma_2^2} \sim F(m-1, n-1) \tag{3.32}$$

In particular, when $\sigma_1^2 = \sigma_2^2$,

$$\frac{s_x^2}{s_y^2} \sim F(m-1, n-1) \tag{3.33}$$

This is the theoretical basis of the F test, used to compare the variances of two sets of data for a significant difference.

The three major sampling distributions cover, respectively, inference about a sample variance, testing of a sample mean, and comparison of two variances. Later sections take up the concrete application of this theory to data analysis.

3.3 Testing for Outliers and Expressing the Measurement Result

If a set of measured data follows a normal distribution exactly, the sample mean and the sample variance can be used directly to estimate the true value and the variance of the population. In an actual experiment, however, an operational mistake or a drift in the conditions often makes an individual value distinctly too large or too small, and such a value is called an outlier. Once these extreme values are removed, the distribution of the remaining data usually lies closer to normal and can be handled by the normal methods described above. Detecting and rejecting outliers is therefore the first step in data processing.

3.3.1 Rejecting Outliers

Many methods can identify an outlier, among them the 3σ rule, the Q test (Dixon's method), Chauvenet's criterion, the Grubbs method, the t test, and the range method. Of these, the International Organization for Standardization (ISO) and the national standards of China both recommend the Grubbs method. The statistic used by this method is

$$G = \frac{|x_{\mathrm{d}} - \bar{x}|}{s} \tag{3.34}$$

where x_{d} is the suspect value, and $\bar{x}$ and s are the mean and the standard deviation of the data set that includes the suspect value.

Using the method requires that a significance level α be set in advance, from which the critical value $G_{\text{crit.}}$ of the G statistic is determined. If the calculated $G > G_{\text{crit.}}$, the value x_{d} is judged to be an outlier and should be rejected. The critical value is calculated as

$$G_{\text{crit.}} = \frac{n-1}{\sqrt{n}} \sqrt{\frac{t^2_{\alpha/(2n),\,n-2}}{n-2+t^2_{\alpha/(2n),\,n-2}}} \tag{3.35}$$

where n is the sample size; α is the significance level, usually taken as $\alpha = 0.05$; and the corresponding confidence level is $P = 1-\alpha$. In practice the critical value of G is obtained from a table (Table 3.6).

Table 3.6: Critical values of $G_{P,n}$

Determinations	Confidence level P		Determinations	Confidence level P	
n	95%	99%	n	95%	99%
3	1.154	1.155	12	2.412	2.636
4	1.481	1.496	13	2.462	2.699
5	1.715	1.764	14	2.507	2.755
6	1.887	1.973	15	2.548	2.806
7	2.020	2.139	16	2.586	2.852
8	2.127	2.274	17	2.620	2.894
9	2.215	2.387	18	2.652	2.932
10	2.290	2.482	19	2.681	2.968
11	2.355	2.564	20	2.708	3.001

Example 3.2 A NaOH solution was determined 6 times in parallel, with the results 0.1049 mol/L, 0.1050 mol/L, 0.1042 mol/L, 0.1086 mol/L, 0.1063 mol/L, and 0.1064 mol/L. Decide whether a suspect value is present in this data set. Take the confidence level as $P = 95\%$.

Solution: the calculated values are

$$\bar{x} = 0.1059\,\text{mol/L}, \quad s = 0.0016\,\text{mol/L}$$

If a suspect value is present, the most likely candidates are 0.1042 mol/L and 0.1086 mol/L, whose G values are

$$G_{0.1042\,\text{mol/L}} = \frac{0.1059\,\text{mol/L} - 0.1042\,\text{mol/L}}{0.0016\,\text{mol/L}} = 1.06$$
$$G_{0.1086\,\text{mol/L}} = \frac{0.1086\,\text{mol/L} - 0.1059\,\text{mol/L}}{0.0016\,\text{mol/L}} = 1.69$$

From the table, $G_{0.95,6} = 1.887$, so the data set contains no outlier.

The Grubbs method rests on the sample mean and the sample standard deviation, and so is comparatively accurate. Rejecting an outlier, however, means recalculating the mean and the standard deviation again and again, which is tedious by hand. Appendix B provides a function, del_outliers_grubbs.m, for calculating and rejecting outliers.

Stop and Think

In the example above both G values lie below the critical value, which means that neither 0.1042 nor 0.1086 is an outlier. Judged by eye alone, though, would you not say that 0.1086 "looks a little large"? That is why a statistical test is needed: intuition cannot settle the question; G and the critical value can.

3.3.2 Expressing the Measurement Result

Once the outliers have been rejected, a set of measured data may be regarded as approximately normally distributed. The sample mean then serves as an unbiased estimate of the true value, and the sample standard deviation as an unbiased

estimate of the population standard deviation. A common practice is to characterize the measurement result and its precision directly by these two statistics:

$$\bar{x}(s, n) \tag{3.36}$$

where $\bar{x}$, s, and n are the sample mean, the sample standard deviation, and the sample size.

Simple and convenient as this practice is, and widely used as it remains, the information it conveys is incomplete. Interval estimation in statistics handles the problem well. Interval estimation gives, at a stated confidence level, the range of values that a statistic may take. The confidence interval for the true value may be written

$$\bar{x} - t_{\alpha,\nu} \times \frac{s}{\sqrt{n}} \leq \mu \leq \bar{x} + t_{\alpha,\nu} \times \frac{s}{\sqrt{n}} \tag{3.37}$$

where α is the two-sided significance level and $\nu = n - 1$ is the number of degrees of freedom.

The practice now recommended is to express a measurement result by the mean together with its uncertainty. The uncertainty of the mean $\bar{x}$ itself is its standard deviation $s/\sqrt{n}$, called the standard uncertainty, which is the standard error; it is denoted by u. The value u corresponds to a confidence level of about 68% and reflects how widely the estimate of the mean is dispersed.

If the confidence level is to be raised to 95% or higher, u is enlarged by a factor, and the result is called the expanded uncertainty, denoted by U:

$$U = ku \tag{3.38}$$

where k is called the confidence factor.

When the number of measurements is large enough, k is determined by the normal distribution (Table 3.7); when the number is small, k is taken as the critical value of the t distribution, $t_{\alpha,\nu}$, where $\alpha = 1 - P$ and $\nu = n - 1$.

Table 3.7: Correspondence between the confidence level P and the confidence factor k under the normal distribution

P	0.50	0.6827	0.90	0.95	0.9545	0.99	0.9973
k	0.675	1	1.645	1.960	2	2.576	3

The measurement result is then expressed as

$$\bar{x} \pm U \tag{3.39}$$

When an expanded uncertainty is reported, the values of U, u, the degrees of freedom ν, the confidence level P, and the confidence factor k should be given at the same time.

Example 3.3 The copper content of a copper alloy was measured 10 times, giving the following values (%): 69.95, 71.12, 65.41, 70.26, 69.63, 69.91, 68.66, 69.26, 68.76, and 69.21. Express the measurement result properly.

Solution: first test the data set for an outlier, which would be either the smallest value, 65.41%, or the largest, 71.12%. The mean of the data set is $\bar{x} = 69.22\%$ and the standard deviation is $s = 1.52\%$. The G statistics are

$$G_{65.41\%} = \frac{69.22\% - 65.41\%}{1.52\%} = 2.51$$

$$G_{71.12\%} = \frac{71.12\% - 69.22\%}{1.52\%} = 1.25$$

Because $G_{65.41\%} > G_{0.95,10} = 2.290$, the value 65.41% is judged to be an outlier and is rejected. The remaining data were tested again and no further outlier was found. From the remaining data,

$$\bar{x} = 69.64\%, \quad s = 0.78\%$$

The standard uncertainty is

$$u = \frac{s}{\sqrt{n}} = \frac{0.78\%}{\sqrt{9}} = 0.26\%$$

At the confidence level $P = 95\%$, the number of measurements is small (degrees of freedom $\nu = 8$), so the confidence factor is taken as $k = t_{0.05,8} = 2.31$, and the expanded uncertainty is

$$U = ku = 2.31 \times 0.26\% = 0.60\%$$

The measurement result for the copper content of the copper alloy is expressed as

$$(69.64 \pm 0.60)\%$$

Common Pitfall

Reporting $\bar{x}(s, n)$ alone is not enough. A complete measurement result should contain two parts, a **best estimate** and an **uncertainty**, usually written as $\bar{x} \pm U$, with the values of U, u, ν, P, and the factor k noted alongside. If any one of these parameters is missing, the reader cannot judge the reliability of the result independently.

3.4 Propagation of Measurement Error

A mass can be weighed directly and a volume can be read directly, but a concentration must be calculated from measured quantities such as mass and volume. Every measured value carries error, and that error is propagated through the calculation into the final result. This section treats the mathematical rules of error propagation.

3.4.1 General Principles

Suppose a quantitative analysis involves m measured quantities X_1, X_2, $\cdots$, X_m, and the quantity finally sought is Y, related to them by

$$Y = g(X_1, X_2, \cdots, X_m) \tag{3.40}$$

If the true values of the measured quantities are μ_1, μ_2, $\cdots$, μ_m, a Taylor expansion about the true values with first-order terms retained gives

$$Y = g(\mu_1, \mu_2, \cdots, \mu_m) + \sum_{i=1}^{m} \left(\frac{\partial g}{\partial X_i}\right)\Big|_{\mu_1,\mu_2,\cdots,\mu_m} (X_i - \mu_i) \tag{3.41}$$

For convenience, write

$$a_0 = g(\mu_1, \mu_2, \cdots, \mu_m), \quad a_i = \left(\frac{\partial g}{\partial X_i}\right)\Big|_{\mu_1,\mu_2,\cdots,\mu_m} \tag{3.42}$$

The expected value of Y is

$$\begin{aligned} \mu_Y &= E[Y] \\ &= E\left[a_0 + \sum_{i=1}^{m} a_i(X_i - \mu_i)\right] \\ &= a_0 + \sum_{i=1}^{m} a_i E[X_i] - \sum_{i=1}^{m} a_i \mu_i \\ &= a_0 \end{aligned} \tag{3.43}$$

The expected value of $(Y - \mu_Y)^2$ is

$$\begin{aligned} \sigma_Y^2 &= E\left[(Y - \mu_Y)^2\right] \\ &= E\left[\left(\sum_{i=1}^{m} a_i(X_i - \mu_i)\right)^2\right] \\ &= E\left[\left(\sum_{i=1}^{m} a_i(X_i - \mu_i)\right)\left(\sum_{j=1}^{m} a_j(X_j - \mu_j)\right)\right] \end{aligned}$$

$$= E\left[\sum_{i=1}^{m} a_i^2 (X_i - \mu_i)^2 + \sum_{i=1}^{m} \sum_{j=1, j\neq i}^{m} a_i a_j (X_i - \mu_i)(X_j - \mu_j)\right]$$
$$= \sum_{i=1}^{m} a_i^2 E\left[(X_i - \mu_i)^2\right] + \sum_{i=1}^{m} \sum_{j=1, j\neq i}^{m} a_i a_j E\left[(X_i - \mu_i)(X_j - \mu_j)\right]$$
$$= \sum_{i=1}^{m} a_i^2 \sigma_i^2 + \sum_{i=1}^{m} \sum_{j=1, j\neq i}^{m} a_i a_j \sigma_{ij} \tag{3.44}$$

When the X_i ($i = 1, 2, \cdots, m$) are uncorrelated with one another, $\sigma_{ij} = 0$, and the expression above reduces to

$$\sigma_Y^2 = \sum_{i=1}^{m} a_i^2 \sigma_i^2 \tag{3.45}$$

If the standard uncertainty of each measured quantity X_i is u_i, and the X_i are mutually uncorrelated, the combined variance of Y is

$$u_Y^2 = \sum_{i=1}^{m} a_i^2 u_i^2 \tag{3.46}$$

and the combined standard uncertainty of Y is

$$u_Y = \sqrt{\sum_{i=1}^{m} a_i^2 u_i^2} \tag{3.47}$$

If the degrees of freedom associated with the individual components are ν_i, the effective degrees of freedom of the combined standard uncertainty are

$$\nu_{\text{eff}} = \frac{u_Y^4}{\sum_{i=1}^{m} \frac{a_i^4 u_i^4}{\nu_i}} \tag{3.48}$$

The effective degrees of freedom is the number of degrees of freedom of the t distribution to which the combined standard uncertainty u_Y is equivalent. Once ν_{eff} is known, the confidence factor k follows from the t distribution. When ν_{eff} is sufficiently large, $k = 2$ (corresponding to a confidence level of about 95%) or

$k = 3$ (about 99.7%) may be used directly. The values $k = 2$ and 3 are approximations, however: if ν_{eff} is small (say $\nu_{\text{eff}} < 10$), the actual confidence level may depart appreciably from the intended one, and k should then be read strictly from a table of the t distribution. A formal report on the evaluation of measurement uncertainty should state clearly the basis for the value of k. This book does not treat the complete procedure for evaluating uncertainty; interested readers may consult the relevant standards and monographs.

The discussion turns next to error propagation in addition and subtraction and in multiplication and division, which also lays the foundation for evaluating uncertainty.

3.4.2 Addition and Subtraction

Let Y be the result of adding and subtracting three measured values A, B, and C, calculated as

$$Y = g(A, B, C) = A + bB - C \tag{3.49}$$

Since

$$\frac{\partial g}{\partial A} = 1, \quad \frac{\partial g}{\partial B} = b, \quad \frac{\partial g}{\partial C} = -1 \tag{3.50}$$

it follows that

$$\sigma_Y^2 = \sigma_A^2 + b^2\sigma_B^2 + \sigma_C^2 \tag{3.51}$$

Example 3.4 A burette used in titrimetric analysis usually carries a stated accuracy tolerance, the largest permissible error of a single reading as given by the manufacturer. If, as an engineering approximation, that tolerance is taken directly as the standard deviation s of a single reading, what is the standard deviation of one completed titration?

Solution: A completed titration requires two readings from the burette, and their difference is the volume delivered. Let the two readings be V_1 and V_2; the delivered volume V is

$$V = g(V_1, V_2) = V_2 - V_1$$

Since

$$\frac{\partial g}{\partial V_1} = -1, \quad \frac{\partial g}{\partial V_2} = 1$$

it follows that

$$s_V = \sqrt{(-s)^2 + s^2} = \sqrt{2}s$$

3.4.3 Multiplication and Division

Let Y be the result of multiplying and dividing three measured values A, B, and C, calculated as

$$Y = g(A, B, C) = \frac{AB}{C} \tag{3.52}$$

Hence

$$\frac{\partial g}{\partial A} = \frac{B}{C}, \quad \frac{\partial g}{\partial B} = \frac{A}{C}, \quad \frac{\partial g}{\partial C} = -\frac{AB}{C^2} \tag{3.53}$$

which gives

$$\begin{aligned} \sigma_Y^2 &= \left(\frac{\partial g}{\partial A}\right)^2 \sigma_A^2 + \left(\frac{\partial g}{\partial B}\right)^2 \sigma_B^2 + \left(\frac{\partial g}{\partial C}\right)^2 \sigma_C^2 \\ &= \left(\frac{B}{C}\right)^2 \sigma_A^2 + \left(\frac{A}{C}\right)^2 \sigma_B^2 + \left(\frac{AB}{C^2}\right)^2 \sigma_C^2 \end{aligned} \tag{3.54}$$

Rearranging gives

$$\frac{\sigma_Y^2}{Y^2} = \frac{\sigma_A^2}{A^2} + \frac{\sigma_B^2}{B^2} + \frac{\sigma_C^2}{C^2} \tag{3.55}$$

A familiar standardization shows how error propagation for multiplication and division is applied.

Example 3.5 A 25.00 mL portion of NaOH solution is transferred with a pipet and titrated with 0.1000 mol/L standard HCl solution, 30.00 mL being consumed. The standard deviation of measuring out solution with the pipet is known to be $s_1 = 0.02$ mL, and the standard deviation of each reading of the burette is $s_2 = 0.01$ mL. Assuming the concentration of the HCl solution to be exact, calculate the standard deviation of the standardization of the NaOH solution.

Solution: the concentration of the NaOH solution is

$$\begin{aligned} c(\mathrm{NaOH}) &= \frac{c(\mathrm{HCl})V(\mathrm{HCl})}{V(\mathrm{NaOH})} \\ &= \frac{0.1000\,\mathrm{mol/L} \times 30.00\,\mathrm{mL}}{25.00\,\mathrm{mL}} \\ &= 0.1200\,\mathrm{mol/L} \end{aligned}$$

The pipet is read once, so the standard deviation of its reading is simply s_1. The burette is read twice, so the standard deviation of the delivered volume is $\sqrt{2}s_2$.

The standard deviation of the standardization of the NaOH solution is calculated as

$$\frac{s^2(\mathrm{NaOH})}{c^2(\mathrm{NaOH})} = \frac{s_1^2}{V^2(\mathrm{NaOH})} + \frac{2s_2^2}{V^2(\mathrm{HCl})}$$

Hence

$$\begin{aligned} s(\mathrm{NaOH}) &= 0.1200\,\mathrm{mol/L} \times \sqrt{\left(\frac{0.02\,\mathrm{mL}}{25.00\,\mathrm{mL}}\right)^2 + 2\left(\frac{0.01\,\mathrm{mL}}{30.00\,\mathrm{mL}}\right)^2} \\ &= 0.0001\,\mathrm{mol/L} \end{aligned}$$

3.4.4 Other Operations

The rules of error propagation for other operations follow from the same reasoning, and several commonly used formulas are collected directly in Table 3.8.

Table 3.8: Several error propagation formulas

Operation	s_Y^2
$Y = \ln A$	$\frac{1}{A^2} s_A^2$
$Y = e^A$	$Y^2 s_A^2$
$Y = A^k$	$k^2 \frac{Y^2}{A^2} s_A^2$

Key Concept

The core principle of error propagation can be stated in one sentence: the uncertainty of an indirectly measured quantity is obtained from the uncertainties of the directly measured quantities by weighting each with the square of its partial derivative and taking the square root of the weighted sum of squares. Addition and subtraction, and multiplication and division, are the two situations met most often in routine analysis, and these two cases suffice for most calculations. More complex problems of propagation can be pursued in the standard documents.

3.5 Significant Figures and the Rules for Their Computation

Section 3.1 introduced the concept of significant figures from the standpoint of reading an instrument: in complete reading we record every digit the instrument displays (for example 1.0610 g, five digits), and in estimated reading we record the certain digits plus one estimated digit (for example 44.5 mL, three digits). Significant figures are all the digits that a single actual measurement can yield; they consist of the certain part plus the final estimated digit. Every significant digit reflects both the resolving power of the instrument and the precision

of the measurement at hand, so adding or dropping digits at will amounts to misreporting the data.

The number of significant figures depends on the precision of the measuring instrument. Consider an analytical balance with a smallest graduation of 0.1 mg, one ten-thousandth of a gram; a weighed value must therefore be read to the fourth decimal place. That last digit fluctuates from reading to reading, which is the random error of the weighing made visible, and it must be recorded exactly as it stands, never dropped. Suppose one weighing gives 0.2152 g. The estimated error of the last digit is ±0.0001 g, the true value lies within 0.2152 ± 0.0001 g, and the total spread is 0.0002 g, the difference between +0.0001 g and −0.0001 g. The largest relative error that may then be introduced is

$$\frac{\pm 0.0001\,\text{g}}{0.2152\,\text{g}} \times 100\% = \pm 0.05\% \tag{3.56}$$

If the balance reading were recorded as 0.215 g alone, the estimated error of the last digit would grow to ±0.001 g and the total spread to 0.002 g, so the largest relative error that may be introduced becomes

$$\frac{\pm 0.001\,\text{g}}{0.215\,\text{g}} \times 100\% = \pm 0.5\% \tag{3.57}$$

The more significant figures, then, the more accurate the result. Digits that go beyond what the accuracy of the measurement can support, however, are not merely meaningless; they are conceptually wrong.

3.5.1 Number of Significant Figures

The number of significant figures depends first of all on the accuracy of the instrument, and the last digit of a recorded value is an estimate. When an object is weighed, for instance, a balance reading of 0.2152 g has four significant figures, and a burette reading of 15.36 mL also has four. Some examples follow.

1.0008	43081	five significant figures
0.1000	10.98%	four significant figures
0.0382	1.08×10^{-10}	three significant figures
0.0054	0.40	two significant figures
0.5	0.02%	one significant figure

Counting significant figures, as these examples show, is usually not difficult. The one real difficulty arises when the digit "0" appears in a number, for a "0" plays a different role in different positions. The following points explain the matter in detail:

- When a "0" stands at the end of a number, or between nonzero digits, that "0" is a genuine significant figure and is counted.
- When a "0" stands before all the nonzero digits, that "0" serves only to locate the decimal point and is not counted.
- When the unit is changed, the number of "0" digits changes, and care must be taken that the number of significant figures does not.

The last case is in fact very common. Measured in grams, 0.0100 g has three significant figures; in milligrams it should be written 10.0 mg, and in micrograms $1.00 \times 10^4\,\mu$g. It must not be written 10000 μg, because the number of significant figures in that form of expression is ambiguous. Given as it stands, 10000 μg would ordinarily be taken to have 1 significant figure. For a number of this kind, one that ends in "0", the position of the decimal point must therefore be marked properly.

Logarithmic quantities such as pH, pM, and log K appear often in quantitative chemical analysis. Their number of significant figures is fixed by the number

of digits in the mantissa, because the integer part corresponds to the power of ten. A pH of 11.20, for instance, has two significant figures in its mantissa; converted to the concentration of H^+ ions, it gives $[H^+] = 6.3 \times 10^{-12}$ mol/L.

Numbers that do not come from measurement, such as multipliers, fractional relationships, π, and e, are not assigned a definite number of significant figures; the number of digits taken is decided by the situation at hand.

In analytical chemistry the following rules are generally used in recording significant figures:

1. When a measurement is recorded, only one doubtful digit is kept, as in the balance reading 0.3281 g, the burette volume reading 28.39 mL, or the pH value 2.87.
2. The result for a component present at more than 10% is generally required to have four significant figures; a component present at 1% $\sim$ 10% requires three; a component present at less than 1% usually requires only two.
3. An error computed from the data is usually given to 1 $\sim$ 2 significant figures, for example a relative error of 0.02% or a standard deviation of 0.23.

That an error keeps 1 $\sim$ 2 significant figures is not an arbitrary choice. Let us first look at the quality of the measurement itself. When the precision is good, the random fluctuation of the data stays within the last digit, and the standard deviation *s* has a magnitude of about one unit in that last place, so it usually carries only one significant figure. When the precision is poorer, the fluctuation spills past the last digit and *s* grows in magnitude, and a second significant figure becomes necessary. An error quoted to one significant figure therefore tends to

accompany a good measurement, whereas two figures suggest that the precision is not satisfactory.

Consider next the precision of the error itself. An error is an estimate as well, and it carries its own error. Take the sample standard deviation s: it is only an estimate of the population standard deviation σ, and it is not exact in itself. Statistical theory gives a relative error of about $1/\sqrt{2\nu}$ for s as an estimate of σ, where $\nu = n - 1$ is the number of degrees of freedom. With only 5 measured values, $\nu = 4$, and this error is about 35%. If the computed s is 0.23, for example, then the true σ is roughly 0.23 ± 0.08. The first significant figure of 0.23, the "2", still means something, but the second, the "3", is already quite unreliable, and one figure is all that should be kept. When the number of measurements is larger, the relative error of s becomes smaller and the second figure is barely usable, so convention keeps at most two; whatever the number of measurements, a third figure is meaningless.

The Guide to the Expression of Uncertainty in Measurement and the Chinese national metrological technical specification JJF 1059.1 likewise require that a reported expanded uncertainty usually keeps only $1 \sim 2$ significant figures. Retaining extra figures does not make the information more reliable; it gives the false impression that the error has been measured very precisely.

Common Pitfall

In working problems and writing laboratory reports, some students copy the error values displayed by the calculator without a thought, writing a relative error of 0.046512% or a standard deviation of 0.233333. These are instances of false precision. The number of significant figures in an error is fixed by the precision with which the error itself is estimated, and usually only $1 \sim 2$ figures are kept; extra digits are meaningless and reveal that the author has not understood what an error means. The proper procedure is to round the error to $1 \sim 2$ significant figures first, and then to use it in deciding the last place retained in the measured result.

3.5.2 Rules for Rounding Significant Figures

In chemical data processing, measured values with different numbers of significant figures often coexist in a single calculation. Since different numbers of significant figures represent different levels of measurement accuracy, the significant figures must be rounded if the final result is to be meaningful. Rounding significant figures means fixing the correct number of significant figures on the basis of the complete result. The steps are as follows:

- Carry the calculation through and obtain the complete, unrounded result;
- Use the rules for computation with significant figures to determine the last significant figure that should be retained;
- Look at the first digit after that last significant figure, also called the trailing digit, and either discard or round up that trailing digit and the digits after it, according to the round-half-to-even rule, known in Chinese by the mnemonic “four discard, six round up, five makes even”.

The round-half-to-even rule is specified as follows:

- When the trailing digit is 4 or less, that digit and all digits after it are discarded;
- When the trailing digit is 6 or more, the last significant figure is increased by 1 and the trailing digit and all digits after it are discarded;
- When the trailing digit is 5, the following cases apply:
  - if the digit 5 is followed only by 0 and the figure before it is odd, round up;
  - if the digit 5 is followed only by 0 and the figure before it is even, discard;
  - if the 5 is followed by any nonzero digit, round up.

The following examples round the numbers on the left to four significant figures:

0.32554	$\longrightarrow$	0.3255
0.42605	$\longrightarrow$	0.4260
15.4565	$\longrightarrow$	15.46
150.650	$\longrightarrow$	150.6
16.0851	$\longrightarrow$	16.09

3.5.3 Rules for Computation with Significant Figures

It was shown earlier that error propagates along the calculation. The rules for computation with significant figures are a concrete application of that principle: the number of digits kept in the final result is not fixed at will but is decided by the least precise of all the input quantities. The core principle can be summarized

in one line: addition and subtraction rest on absolute error, multiplication and division on relative error.

From Eq. (3.41) of Section 3.4.1, if we define

$$\Delta Y = Y - g(\mu_1, \mu_2, \cdots, \mu_m) \tag{3.58}$$

and

$$\Delta X_i = X_i - \mu_i \tag{3.59}$$

then we obtain:

$$\Delta Y = \sum_{i=1}^{m} \left(\frac{\partial g}{\partial X_i} \right) \Big|_{\mu_1, \mu_2, \cdots, \mu_m} \Delta X_i \tag{3.60}$$

Eq. (3.60) is the common starting point for all the rules of computation that follow. The way in which the errors combine depends on the nature of the partial derivatives. The resulting rules of computation are treated below, for addition and subtraction and for multiplication and division separately.

Addition and Subtraction

Consider a concrete instance, in which a quantity Y is the result of adding and subtracting three measured quantities A, B, and C:

$$Y = A + bB - C \tag{3.61}$$

Since

$$\frac{\partial g}{\partial A} = 1, \quad \frac{\partial g}{\partial B} = b, \quad \frac{\partial g}{\partial C} = -1 \tag{3.62}$$

substitution into Eq. (3.60) gives the absolute error of Y:

$$\Delta Y = \Delta A + b\Delta B - \Delta C \tag{3.63}$$

Eq. (3.63) shows that the final error is decided by the term with the largest absolute error. In addition and subtraction, therefore, the final result should be retained only to the last place of the number with the largest absolute error.

Example 3.6 Find: $135.621 + 0.33 + 21.2163 = ?$

Solution: if the calculation is carried out directly, the result is:

$$R = 135.621 + 0.33 + 21.2163 = 157.1673$$

Since 0.33 has the largest absolute error, ± 0.01, the second decimal place is already a doubtful value, and the computed result should therefore be rounded to the second decimal place:

$$\begin{aligned} R &= 135.621 + 0.33 + 21.2163 \\ &= 135.62 + 0.33 + 21.22 \\ &= 157.17 \end{aligned}$$

Multiplication and Division

When Y is the result of multiplying and dividing three measured quantities A, B, and C:

$$Y = g(A, B, C) = \frac{AB}{C} \tag{3.64}$$

Since

$$\frac{\partial g}{\partial A} = \frac{B}{C}, \quad \frac{\partial g}{\partial B} = \frac{A}{C}, \quad \frac{\partial g}{\partial C} = -\frac{AB}{C^2} \tag{3.65}$$

it follows that:

$$\Delta Y = \frac{B}{C}\Delta A + \frac{A}{C}\Delta B - \frac{AB}{C^2}\Delta C \tag{3.66}$$

Rearranging gives:

$$\frac{\Delta Y}{Y} = \frac{\Delta A}{A} + \frac{\Delta B}{B} - \frac{\Delta C}{C} \tag{3.67}$$

Eq. (3.67) shows that the relative error of the final result is decided by the measured value with the largest relative error. In multiplication and division, therefore, the final result should keep the number of significant figures of that measured value.

Example 3.7 Find: $0.0121 \times 25.64 \times 1.05782 = ?$

Solution: the relative errors of the numbers are as follows:

Number	Relative error
0.0121	$\frac{\pm 0.0001}{0.0121} \times 100\% = \pm 0.8\%$
25.64	$\frac{\pm 0.01}{25.64} \times 100\% = \pm 0.04\%$
1.05782	$\frac{\pm 0.00001}{1.05782} \times 100\% = \pm 0.0009\%$

The number with the largest relative error is 0.0121, which has three significant figures. The computed result should therefore keep three significant figures, as follows:

$$\begin{aligned} R &= 0.0121 \times 25.64 \times 1.05782 \\ &= 0.0121 \times 25.6 \times 1.06 \\ &= 0.328 \end{aligned}$$

In multiplication and division, a number whose first digit is 8 or 9, such as 9.00 or 8.92, appears often. Although such a number has only three significant figures, it lies close to 10.00, so numbers of this kind are usually treated as having four significant figures.

3.5.4 A Convention in Computation with Significant Figures*

This section introduces a particular convention for computation with significant figures, one that applies only to the material presented in this book.

For historical reasons, many physical constants in chemistry, stability constants and solubility products among them, were reported with a number of significant figures limited by the measurement techniques of the time. If these constants were treated strictly by the ordinary rules of significant figures, the final result might keep too few significant figures to reflect the precision of modern analytical measurement. To avoid this problem, the book adopts the following convention:

- The number of significant figures of a chemical parameter, such as a stability constant or an equilibrium constant, does not serve as a basis for rounding in a calculation; in principle the parameter is treated as an exact constant;
- The number of significant figures of some noncritical quantities, such as pH values, likewise does not enter the rounding decision, and these are treated as exact constants in principle unless a specified precision requires otherwise;
- Every directly measured physical quantity, such as a mass or a volume, should normally be taken to no fewer than four significant figures. The significant figures retained in the subsequent calculation rest mainly on the significant figures of these measured quantities.

3.6 Evaluation of Quantitative Analytical Results

The preceding sections have set out how data are treated and how results are expressed in a standardized way. Whether the treated result is reliable still has

to be evaluated. In statistical terms that evaluation is a hypothesis test, that is, the use of data to decide whether a given hypothesis holds. This section focuses on the two questions that arise most often in quantitative analysis. Has a mean departed from the true value? Is there a real difference between two means?

3.6.1 Evaluating a Mean against the True Value

The following relation was introduced together with the t distribution:

$$t = \frac{\bar{x} - \mu}{s/\sqrt{n}} \sim t(n-1) \tag{3.68}$$

Here $\bar{x}$ is the sample mean, s the sample standard deviation, n the sample size, and μ the population mean.

If the measured values are affected by random error alone, then $\bar{x}$ is an unbiased estimate of the population true value μ, and the statistic t defined by Equation 3.68 should follow the t distribution. Using that equation to test how far the sample mean deviates from the true value is the t test.

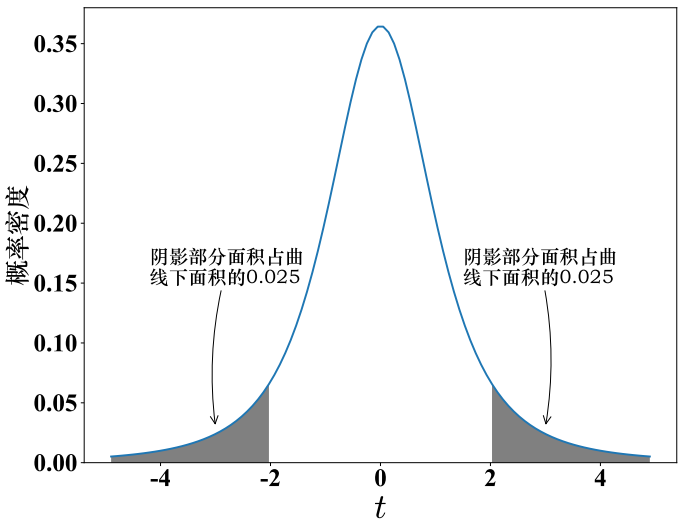


Figure 3.11: Schematic diagram of the t test.

Figure 3.11 is a schematic diagram of the t test. The shaded area accounts for 5% of the total probability (2.5% at each tail), and the central region accounts for 95%. If the calculated value of t falls in the central region, the event is a high-probability one. On this basis the 95% confidence interval of t can be marked out, its boundaries being the critical values $t_{\alpha,\nu}$. A value of t inside the critical values is taken to show no significant difference between the sample mean and the true value; a value beyond them, to show that such a difference exists. What is under test is "whether the mean deviates from the true value," and the direction of the deviation (high or low) is unknown in advance, so a two-sided test is shown here, the two tails in the figure corresponding to the risk of an upward deviation and of a downward deviation respectively.

This brings up a key choice in the t test: **one-sided or two-sided?** The two differ in the form of the alternative hypothesis. If we care only whether the mean lies beyond a particular direction, as in "is the result of the new method significantly high," a one-sided test is used, the rejection region is concentrated in one tail of the distribution, and the whole of the risk α is placed in that tail. If we care about deviation of the mean in either direction, as in "does the result of the new method differ from the standard value," a two-sided test is used, the risk α is divided equally between the two tails, $\alpha/2$ to each. The great majority of tests on means in analytical chemistry are two-sided, for instance in judging whether a new method introduces systematic error or whether two methods differ, because the direction of the deviation cannot be foreseen before the test. In the corresponding critical value $t_{\alpha,\nu}$, α denotes the significance level of a two-sided test rather than a one-sided one, and in consulting a table of the t distribution one should note whether the heading gives a one-sided or a two-sided value.

To state this testing logic fully in statistical language, a pair of mutually ex-

clusive hypotheses must be set up:

- **Null hypothesis** (denoted H_0): there is **no** difference between the mean and the true value, that is, $\mu = \mu_0$, where μ_0 is a known standard value, such as the certified value of a reference material, the value given by a standard method, or a limit prescribed by a standard. In the context of the preceding discussion, H_0 means that "the measured data are affected only by random error, and no systematic error has been introduced." The word "null" means "empty," indicating that no source of difference exists apart from the known random error.
- **Alternative hypothesis** (denoted H_1): a difference **does** exist between the mean and the true value, that is, $\mu \neq \mu_0$. It corresponds to "systematic error is present in the measured data."

The true value μ is unknown, so Eq. (3.28) cannot be evaluated directly. Substituting the relation $\mu = \mu_0$ asserted by the null hypothesis gives the test statistic

$$t = \frac{\bar{x} - \mu_0}{s/\sqrt{n}} \tag{3.69}$$

The task of the t test is then to judge, from the sample data, whether there is sufficient evidence to reject H_0. If $|t| > t_{\alpha,\nu}$, the probability of obtaining the present value of t under the assumption that the null hypothesis holds is below α, so the event is a low-probability one; we therefore reject H_0, accept H_1, and conclude that "a significant difference exists." If instead $|t| \leq t_{\alpha,\nu}$, the available sample is not yet sufficient to overturn H_0, and the conclusion is "no significant difference." This framework applies not only to the comparison of means; it is also the common basis of the two-sample test that follows and of significance tests in regression.

"Failure to reject H_0" is not the same as "H_0 is true." When $|t| \leq t_{\alpha,\nu}$, all that can be said is that "at the present sample size and significance level there is

not enough evidence to prove a difference." Absolute statements such as "the two were proved identical" or "the new method is free of error" should not be used. The outcome may arise either because there really is no difference, or because the sample is too small and the measurement not precise enough to detect a difference that does exist; the latter case is exactly what the type II error describes, and it is treated in Section 3.6.3.

In the critical value $t_{\alpha,\nu}$, α is the significance level, and $\alpha = 0.05$ corresponds to a confidence level of 95%; ν is the degrees of freedom, equal to the sample size minus 1, that is, $\nu = n - 1$. If $|t| > t_{\alpha,\nu}$, a statistically significant difference exists between the mean and the true value; if $|t| < t_{\alpha,\nu}$, there is no significant difference.

Example 3.8 A new method was used to determine the mass fraction of aluminum in a primary standard alum, and the following 9 results were obtained: 10.74%, 10.77%, 10.77%, 10.77%, 10.81%, 10.82%, 10.73%, 10.86%, and 10.81%. The standard value for the aluminum content of the alum is known to be 10.77%, that is, $\mu_0 = 10.77\%$. Has the new method introduced systematic error? Take the significance level as 0.05.

Solution: a Grubbs test for outliers is carried out first. The calculation gives

$$\bar{x} = 10.79\%, \quad s = 0.042\%$$

The possible outliers are 10.86% and 10.73%, with G statistics:

$$G_{10.86\%} = \frac{10.86\% - 10.79\%}{0.042\%} = 1.67$$
$$G_{10.73\%} = \frac{10.79\% - 10.73\%}{0.042\%} = 1.43$$

The table gives $G_{0.95,9} = 2.215$. Both are smaller than the critical value, so there are no outliers.

Now the t test, with $\nu = 9 - 1 = 8$:

$$\begin{aligned} t &= \frac{|\bar{x} - \mu_0|}{s} \times \sqrt{n} \\ &= \frac{10.79\% - 10.77\%}{0.042\%} \times \sqrt{9} \\ &= 1.4 \end{aligned}$$

The table gives $t_{0.05,8} = 2.31$. Since $t < t_{0.05,8}$, no significant difference exists between $\bar{x}$ and μ_0; that is, the new method has introduced no systematic error.

Example 3.9 A 10 mL Class A pipet was calibrated 10 times, with the following results: 10.002, 9.993, 9.984, 9.996, 9.989, 9.983, 9.991, 9.990, 9.988, and 9.999. Does this pipet need correction?

Solution: the calculation gives

$$\bar{x} = 9.992\,\text{mL}, \quad s = 0.006\,\text{mL}$$

The Grubbs statistics are

$$G_{9.983\,\text{mL}} = \frac{9.992\,\text{mL} - 9.983\,\text{mL}}{0.006\,\text{mL}} = 1.5$$

$$G_{10.002\,\text{mL}} = \frac{10.002\,\text{mL} - 9.992\,\text{mL}}{0.006\,\text{mL}} = 1.7$$

Both are smaller than the critical value, so there are no outliers. The t test gives

$$\begin{aligned} t &= \frac{9.992\,\text{mL} - 10.000\,\text{mL}}{0.006\,\text{mL}/\sqrt{10}} \\ &= 4.2 > t_{0.05,9} = 2.26 \end{aligned}$$

This result shows a significant difference between the mean and the nominal value, so the pipet needs correction. The correction can be made in either of two ways: in later use of the pipet, replace the nominal value 10.00 mL with the measured mean 9.992 mL obtained from the calibration when the concentration is calculated; or leave the nominal value unchanged and apply a volume correction of +0.008 mL to the final result. The two procedures are equivalent, and either one removes the systematic error introduced by the volume bias of the pipet.

Example 3.10 A water quality monitoring station determined the heavy metal content of a river. Under the national regulation, the cadmium content of the river must not exceed 0.0050 mg/L. The monitoring staff collected 8 water samples and found a mean cadmium content of 0.0054 mg/L with a standard deviation of 0.0003 mg/L. Does the cadmium content of the river exceed the limit? Take the significance level as 0.05.

Solution: only one direction matters in this example, whether the limit has been exceeded, so a **one-sided test** should be used.

Null hypothesis: $\mu \leq 0.0050$ (the limit is not exceeded); alternative hypothesis: $\mu > 0.0050$ (the limit is exceeded).

The t statistic is

$$t = \frac{0.0054 - 0.0050}{0.0003/\sqrt{8}} = 3.77$$

The table of the t distribution gives 1.895 as the critical value for a one-sided test ($\nu = 7$, one-sided $\alpha = 0.05$). Since $t = 3.77 > 1.895$, the null hypothesis is rejected, and we may be 95% confident that the cadmium content of the

river already exceeds the limit.

For comparison: had a two-sided test been used by mistake, the critical value would have been $t_{0.05,7} = 2.365$. The conclusion would still be that the limit is exceeded, but the larger critical value lowers the power of the test. In analytical chemistry, therefore, when the direction of interest is clear, as in "is the content below the detection limit" or "is the recovery significantly below 100%," a one-sided test should be chosen for higher power.

3.6.2 Evaluating Two Means

Comparing two means is important in analytical chemistry. When a new quantitative method is developed, for instance, its results are often compared with those of a standard method to judge whether it has the potential to replace the established one. The procedure runs as follows: (1) determine the same sample separately by the new method and by the standard method, obtaining two sets of data; (2) calculate the sample mean and the sample standard deviation of each; (3) apply the t test to the two means. If no statistical difference appears at the given significance level, the new method shows no systematic bias against the standard method and may be considered a replacement; otherwise the two methods are not interchangeable.

The t statistic for comparing two means is the two-sample t statistic given by Equation 3.30. For convenience in calculation, $\sqrt{1/m + 1/n}$ has here been absorbed into the denominator, giving the form

$$t = \frac{(\bar{x}_1 - \bar{x}_2)}{s_{\bar{x}_1 - \bar{x}_2}} \tag{3.70}$$

Here $\bar{x}_1$ and $\bar{x}_2$ are the means obtained by the two methods, and $s_{\bar{x}_1 - \bar{x}_2}$ is the

standard error of the difference, calculated as

$$s_{\bar{x}_1-\bar{x}_2} = \sqrt{\frac{(m-1)s_1^2 + (n-1)s_2^2}{m+n-2}\left(\frac{1}{m}+\frac{1}{n}\right)} \tag{3.71}$$

where s_1 and s_2 are the sample standard deviations of the two groups. The test has $m + n - 2$ degrees of freedom.

Before Equation 3.70 is applied, an F test is needed to judge whether the population variances represented by the two samples are homogeneous, that is, whether σ_1^2 and σ_2^2 differ significantly. The test rests on the ratio of the two sample variances s_1^2 and s_2^2. Only when the test shows no significant difference between the two population variances may Equation 3.71 be used to calculate the standard error of the difference; otherwise a corrected formula is required, which is not pursued here.

Example 3.11 The mass fraction of niobium in an alloy was determined by two different methods, with the following results:

Method 1	1.21%	1.23%	1.24%	1.24%	1.25%	1.27%
Method 2	1.30%	1.32%	1.33%	1.33%	1.34%	1.36%

Is there a significant difference between the two methods? Take the significance level as 0.05.

Solution: a Grubbs test is first applied to each of the two sets of data, and neither is found to contain an outlier.

Method 1: $\bar{x}_1 = 1.24\%$, $s_1 = 0.020\%$. The largest deviation is

$$|1.21\% - 1.24\%| = |1.27\% - 1.24\%| = 0.03\%$$

$$G = 0.03\%/0.020\% = 1.5 < G_{0.95,6} = 1.887$$

No outliers.

Method 2: $\bar{x}_2 = 1.33\%$, $s_2 = 0.020\%$. The largest deviation is

$$|1.30\% - 1.33\%| = |1.36\% - 1.33\%| = 0.03\%$$

$$G = 1.5 < 1.887$$

No outliers.

Neither set contains an outlier, and $n_1 = n_2 = 6$. An F test is performed to judge whether the two population variances are homogeneous:

$$F = \frac{(0.020\%)^2}{(0.020\%)^2} = 1 < F_{0.05,5,5}$$

The two variances do not differ significantly, so the standard error of the difference can be calculated:

$$\begin{aligned} s_{\bar{x}_1 - \bar{x}_2} &= \sqrt{\frac{(n_1 - 1)s_1^2 + (n_2 - 1)s_2^2}{n_1 + n_2 - 2}\left(\frac{1}{n_1} + \frac{1}{n_2}\right)} \\ &= \sqrt{\frac{5 \times (0.020\%)^2 + 5 \times (0.020\%)^2}{6 + 6 - 2}\left(\frac{1}{6} + \frac{1}{6}\right)} \\ &= 0.012\% \end{aligned}$$

The t statistic is

$$t = \frac{\bar{x}_1 - \bar{x}_2}{s_{\bar{x}_1 - \bar{x}_2}} = \frac{1.33\% - 1.24\%}{0.012\%} = 7.5$$

Since $t = 7.5 > t_{0.05,10} = 2.23$, a significant difference exists between the two analytical methods.

3.6.3 Type I and Type II Errors in Hypothesis Testing

The t test examples above all took $\alpha = 0.05$ as the boundary for deciding "whether a significant difference exists." What is the statistical meaning of α? It

bears directly on the two possible errors in a hypothesis test.

A **type I error** is the rejection of H_0 when it is in fact true, that is, "detecting a difference where none exists." Its probability of occurrence is exactly the significance level α. If $\alpha = 0.05$, then even when H_0 holds there is still a 5% probability of reaching the mistaken conclusion "a significant difference exists" because of random fluctuation.

A **type II error** is the failure to reject H_0 when it is in fact false, that is, "failing to detect a difference that exists." Its probability is denoted β. The size of β depends on the sample size n, on the size of the true deviation, and on the chosen α; it cannot be set directly by the analyst.

The probabilities of the two errors trade off against each other: lowering α reduces the type I error but raises the risk of a type II error, and the reverse is equally true. In analytical chemistry the type I error is usually the greater concern, and $\alpha = 0.05$ or 0.01 is customary.

3.6.4 One-Way Analysis of Variance*

The t test was used above to compare two means. When three or more means must be compared at the same time, however, carrying out t tests pair by pair accumulates type I error: for k groups, $\binom{k}{2}$ pairwise tests are required, and even if all the group means are in fact equal, the probability of at least one "false significance" is far higher than the nominal α. With $k = 3$, for example, 3 tests are needed and the overall risk of a mistaken judgment is about $1-(1-0.05)^3 \approx 14\%$. The correct tool for comparing several means at once is analysis of variance (ANOVA).

The basic idea of analysis of variance is to split the total variation of all the observations into two parts: one arising from differences among the groups, the other from random fluctuation within the groups. If the between-group variation

is significantly larger than the within-group variation, the group means do differ.

Suppose there are k groups, the ith group containing n_i observations $x_{i1}, x_{i2}, \cdots, x_{in_i}$, with $N = \sum_{i=1}^{k} n_i$ observations in all. Let the mean of the ith group be $\bar{x}_i$ and the grand mean be $\bar{x}$. Define the total sum of squares, the between-group sum of squares, and the within-group sum of squares:

$$S_T = \sum_{i=1}^{k}\sum_{j=1}^{n_i}(x_{ij} - \bar{x})^2, \quad S_A = \sum_{i=1}^{k} n_i(\bar{x}_i - \bar{x})^2, \quad S_e = \sum_{i=1}^{k}\sum_{j=1}^{n_i}(x_{ij} - \bar{x}_i)^2 \tag{3.72}$$

The three satisfy $S_T = S_A + S_e$. S_A has $k-1$ degrees of freedom and S_e has $N-k$. Define the mean squares:

$$MS_A = \frac{S_A}{k-1}, \qquad MS_e = \frac{S_e}{N-k} \tag{3.73}$$

The null hypothesis H_0 is that all the group means are equal. When the groups follow normal distributions with a common variance and H_0 holds, the statistic

$$F = \frac{MS_A}{MS_e} \sim F(k-1, N-k) \tag{3.74}$$

If $F > F_{\alpha,k-1,N-k}$, H_0 is rejected, and the group means are judged to differ significantly.

Example 3.12 Three analytical methods (labeled A, B, and C) were each applied in 4 replicate determinations of the calcium content (mg/L) of the same standard solution; the results follow. Do the results of the three methods differ significantly? Take the significance level as 0.05.

Method	Measured value (mg/L)			
A	40.1	40.3	40.2	40.2
B	40.8	40.9	40.7	40.8
C	40.2	40.0	40.1	40.1

Solution: $k = 3$, $n_1 = n_2 = n_3 = 4$, $N = 12$. The group means and the grand mean are

$$\bar{x}_A = 40.2, \quad \bar{x}_B = 40.8, \quad \bar{x}_C = 40.1$$

$$\bar{x} = \frac{40.2 \times 4 + 40.8 \times 4 + 40.1 \times 4}{12} = 40.367$$

Between-group sum of squares:

$$S_A = 4(40.2 - 40.367)^2 + 4(40.8 - 40.367)^2 + 4(40.1 - 40.367)^2 = 1.147$$

Within-group sum of squares:

$$S_e = \sum_{j=1}^{4}(x_{Aj} - 40.2)^2 + \sum_{j=1}^{4}(x_{Bj} - 40.8)^2 + \sum_{j=1}^{4}(x_{Cj} - 40.1)^2 = 0.060$$

Hence

$$F = \frac{S_A/(k-1)}{S_e/(N-k)} = \frac{1.147/2}{0.060/9} = 86.0$$

The table of the F distribution gives $F_{0.05,2,9} = 4.26$. Since $F = 86 > F_{0.05,2,9}$, the null hypothesis is rejected, and the results of the three methods are judged to differ significantly.

Common Pitfall

Analysis of variance can tell only whether the group means are not all equal; it cannot point to which two groups differ. To pin down where the difference lies, a multiple comparison (the Tukey method, for example) must be carried out after a significant result in the analysis of variance, and that is not pursued here.

3.7 Linear Correlation and Regression

In analytical chemistry the response signal of an instrument is often linearly related to the concentration of the analyte: absorbance is proportional to concentration, peak area to the amount injected, and so on. A series of standard solutions establishes that straight line, and the concentration is read back from the signal measured for the sample. This procedure is known as the calibration curve method. This section discusses how the method of least squares fits that line, how the significance of the regression equation is tested, and how the equation is used for prediction.

3.7.1 The Mathematical Model

Suppose two measured quantities y and x stand in the following linear relation:

$$y = \beta_1 x + \beta_0 + \varepsilon \tag{3.75}$$

Here β_1 is the slope of the line, β_0 its intercept, and ε the random error.

The meaning of β_1 changes with the application. If y is the response of an instrument, β_1 stands for the sensitivity of that instrument. In quantitative analysis β_0 usually comes from a systematic background signal, such as the reagent blank. The term ε is the random error of the measurement process, and it is normally assumed to follow the normal distribution $N(0, \sigma^2)$.

In an actual measurement x may also carry error. To keep the model tractable, however, the error in x is generally assumed to be far smaller than that in y and is therefore neglected.

Equation 3.75 is the mathematical model for the correlation analysis between the two measured quantities y and x. To obtain the concrete form of that

model, the values of β_1 and β_0 must be estimated from the actual measurement data. Several estimation methods are available, and the most basic and most widely used of them is simple linear regression.

3.7.2 Simple Linear Regression

Measurement yields a set of values for x, namely $x_1, x_2, \cdots, x_n$, and correspondingly a set of values for y, namely $y_1, y_2, \cdots, y_n$. In examining the relation between the two sets, a scatter plot is usually drawn first for a preliminary judgment, as Figure 3.12 shows. The data points in the figure do not fall exactly on a straight line, yet they display a linear trend.

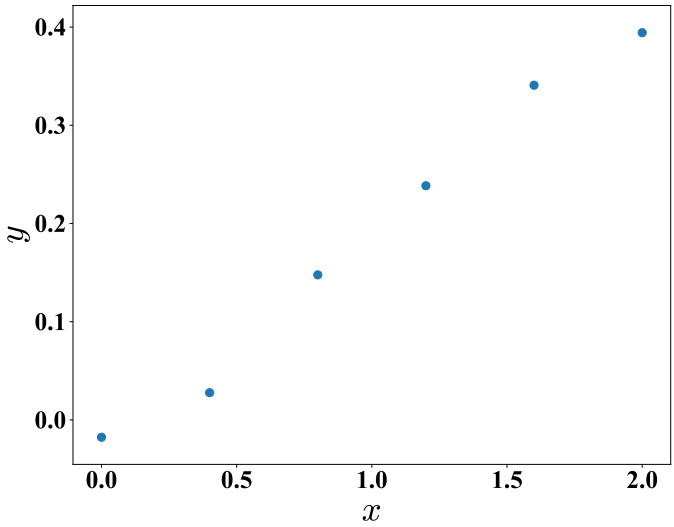


Figure 3.12: Scatter plot of a set of measured data.

What keeps the measured points off a single straight line is chiefly that the random error ε differs from one point to the next. The random error ε likewise has a set of values, $\varepsilon_1, \varepsilon_2, \cdots, \varepsilon_n$; it is only that we do not know the value of any individual ε. Every data point nevertheless satisfies Equation 3.75, that is:

$$y_i = \beta_1 x_i + \beta_0 + \varepsilon_i, \quad i = 1, 2, \cdots, n. \tag{3.76}$$

The subscript i in Equation 3.76 invites misunderstanding. A beginner may suppose that every value of i comes with its own pair (β_1, β_0), so that i running from 1 to n implies n different straight lines. That reading is wrong. In fact β_1 and β_0 are parameters shared by all the data points, and the subscript i merely says that each point obeys the same linear relation. The result therefore cannot be any single line picked from among the n points; what is wanted is the one line that best represents all of them taken together. The least squares principle serves that purpose.

Rewrite Equation 3.76 as

$$\varepsilon_i = y_i - (\beta_1 x_i + \beta_0), \quad i = 1, 2, \dots, n. \tag{3.77}$$

Equation 3.77 writes ε_i as the difference between the measured value and the model value. The model value is given by the true parameters, and since those parameters are unknown, ε_i cannot be computed. In general $y_i - (\beta_1 x_i + \beta_0)$ measures how far the ith data point departs from the line, and it is commonly called the fitted residual. The best-fitting line for the data is obtained by finding the set of parameters (β_1 and β_0) that makes the total residual as small as possible. Using the algebraic sum of the residuals, $\sum_{i=1}^{n} \varepsilon_i$, directly would fail to measure the total error because positive and negative terms cancel; the sum of squared residuals is therefore constructed instead:

$$Q(\beta_1, \beta_0) = \sum_{i=1}^{n} \varepsilon_i^2 = \sum_{i=1}^{n} \Big(y_i - (\beta_1 x_i + \beta_0)\Big)^2 \tag{3.78}$$

The true values of the parameters cannot be obtained, so we seek estimates $\hat{\beta}_1$ and $\hat{\beta}_0$ that make the sum of squared residuals as small as possible, that is, we solve:

$$Q(\hat{\beta}_1, \hat{\beta}_0) = \min_{\beta_0, \beta_1} Q(\beta_1, \beta_0) \tag{3.79}$$

Differentiating Q partially with respect to β_1 and β_0 gives

$$\frac{\partial Q}{\partial \beta_1} = \sum_{i=1}^{n} -2(y_i - \beta_1 x_i - \beta_0)x_i \tag{3.80}$$

$$\frac{\partial Q}{\partial \beta_0} = \sum_{i=1}^{n} -2(y_i - \beta_1 x_i - \beta_0) \tag{3.81}$$

When $\beta_1 = \hat{\beta}_1$ and $\beta_0 = \hat{\beta}_0$, Q reaches its minimum, that is:

$$\left.\frac{\partial Q}{\partial \beta_1}\right|_{\beta_1=\hat{\beta}_1,\beta_0=\hat{\beta}_0} = \sum_{i=1}^{n} -2(y_i - \hat{\beta}_1 x_i - \hat{\beta}_0)x_i = 0 \tag{3.82}$$

$$\left.\frac{\partial Q}{\partial \beta_0}\right|_{\beta_1=\hat{\beta}_1,\beta_0=\hat{\beta}_0} = \sum_{i=1}^{n} -2(y_i - \hat{\beta}_1 x_i - \hat{\beta}_0) = 0 \tag{3.83}$$

Solving gives

$$\hat{\beta}_0 = \frac{\sum_{i=1}^{n} y_i}{n} - \hat{\beta}_1 \frac{\sum_{i=1}^{n} x_i}{n} = \bar{y} - \hat{\beta}_1 \bar{x} \tag{3.84}$$

$$\hat{\beta}_1 = \frac{\sum_{i=1}^{n} y_i x_i - \frac{\left(\sum_{i=1}^{n} x_i\right)\left(\sum_{i=1}^{n} y_i\right)}{n}}{\sum_{i=1}^{n} x_i^2 - \frac{\left(\sum_{i=1}^{n} x_i\right)^2}{n}} \tag{3.85}$$

Provided the model assumption of Equation 3.75 holds, the estimators have the following sampling distributions:

$$\hat{\beta}_0 \sim N\left(\beta_0, \left(\frac{1}{n} + \frac{\bar{x}^2}{\sum_{i=1}^{n}(x_i - \bar{x})^2}\right)\sigma^2\right) \tag{3.86}$$

$$\hat{\beta}_1 \sim N\left(\beta_1, \frac{\sigma^2}{\sum_{i=1}^{n}(x_i - \bar{x})^2}\right) \tag{3.87}$$

For a given $x = x_0$, the predicted value $\hat{y}_0$ of the response variable satisfies

$$\hat{y}_0 = \hat{\beta}_0 + \hat{\beta}_1 x_0 \sim N\left(\beta_0 + \beta_1 x_0, \left(1 + \frac{1}{n} + \frac{(x_0 - \bar{x})^2}{\sum_{i=1}^{n}(x_i - \bar{x})^2}\right)\sigma^2\right) \tag{3.88}$$

3.7.3 Testing the Significance of the Regression Equation

Mathematically, any set of data for y and x will yield a linear equation through Equations 3.84 and 3.85, whether or not a linear relation truly exists between y and x. Once the equation has been established, therefore, it must still be tested for statistical meaning. The statistical tests in common use are the F test, the t test, and the test on the correlation coefficient. In simple linear regression, when the random error ε satisfies the assumptions of normality, independence, and homogeneity of variance, the three are essentially equivalent, so only the F test is discussed here.

Testing the significance of a regression equation by the F test involves several statistics. The first is the total sum of squares, defined as

$$S_T = \sum_{i=1}^{n} (y_i - \bar{y})^2 \tag{3.89}$$

S_T reflects the total degree of variation of the measured values. According to Equation 3.75, that variation should have two sources: variation in y caused by variation in x, and variation in y caused by measurement error. The first is described by the statistic defined below:

$$S_R = \sum_{i=1}^{n} (\hat{y}_i - \bar{y})^2 \tag{3.90}$$

S_R is called the regression sum of squares; it measures how much of the variation in the dependent variable y is explained by variation in the independent variable x. The larger S_R is, the stronger the explanatory power of the regression model over the data.

Complementing S_R is the deviation that generally exists between the predicted value $\hat{y}_i$ and the measured value y_i. This portion of the variation, left unexplained by the regression model, is represented by the sum of squares of resid-

uals S_e, defined as

$$S_e = \sum_{i=1}^{n} (y_i - \hat{y}_i)^2 \tag{3.91}$$

The sum of squares of residuals allows the following statistic to be constructed:

$$\hat{\sigma}^2 = \frac{S_e}{n-2} \tag{3.92}$$

Statistical theory shows that $\hat{\sigma}^2$ is an unbiased estimate of σ^2, so in actual calculation $\hat{\sigma}^2$ may replace σ^2. When $y_1, y_2, \cdots, y_n$ are mutually independent and $y_i \sim N(\beta_0 + \beta_1 x_i, \sigma^2)$, the following relation holds:

$$S_e/\sigma^2 \sim \chi^2(n-2) \tag{3.93}$$

When the null hypothesis $H_0 : \beta_1 = 0$ holds, the variation in y arises entirely from random error, and we have:

$$S_R/\sigma^2 \sim \chi^2(1) \tag{3.94}$$

Equation 3.94 reveals an important fact: if $\beta_1 = 0$, the regression equation reflects nothing but the fluctuation of random error. This is exactly what the "null" in the null hypothesis means, namely that no systematic association between x and y is acknowledged, and that every change in y is attributed to random error. The position is entirely consistent with the H_0 of the t test discussed earlier: the null hypothesis always assumes that no difference exists apart from random error. On this basis the following F statistic can be constructed:

$$F = \frac{S_R}{S_e/(n-2)} \tag{3.95}$$

The value of F is then compared with the critical value $F_{\alpha,1,n-2}$. If $F > F_{\alpha,1,n-2}$, the null hypothesis $\beta_1 = 0$ is rejected, which shows that a linear relation does exist between y and x and that the regression equation is statistically significant.

From the sums of squared deviations above, a further statistic can be defined:

$$R^2 = \frac{S_R}{S_T} = 1 - \frac{S_e}{S_T} \tag{3.96}$$

This statistic gives the proportion of the total deviation in y that can be explained by the effect of x. R^2 is therefore the coefficient of determination, and its square root is called the correlation coefficient. The correlation coefficient is also often used to assess the goodness of fit of a linear regression model. It is commonly held that a correlation coefficient reaching 0.999 shows the regression equation to have a significant linear correlation, meaning that a linear correlation does exist between the two variables y and x. Judging the significance of a regression equation by the size of R^2 alone is not enough, however; the rigorous practice is to verify it further by a statistical test.

A linear regression analysis of the data in Figure 3.12 gives the regression equation shown in Figure 3.13 together with the corresponding straight line. The line lies among the measured points and represents them well. The F test shows the linear relation to be significant.

Stop and Think

A careful reader may ask: if the F test is significant and $R^2 > 0.999$, is the calibration curve "perfect"? Not necessarily. A significant F test says only that a significant linear trend exists between y and x, and a high R^2 says only that the linear relation accounts for most of the variation. Neither guarantees that the data are strictly linear over the whole concentration range, and neither rules out a loss of measurement precision at high concentration (heteroscedasticity). Beyond the F test and R^2, therefore, the distribution of the residuals $\varepsilon_i = y_i - \hat{y}_i$ should also be examined: only when the residuals scatter randomly above and below the zero line with no obvious trend is the model more reliable.

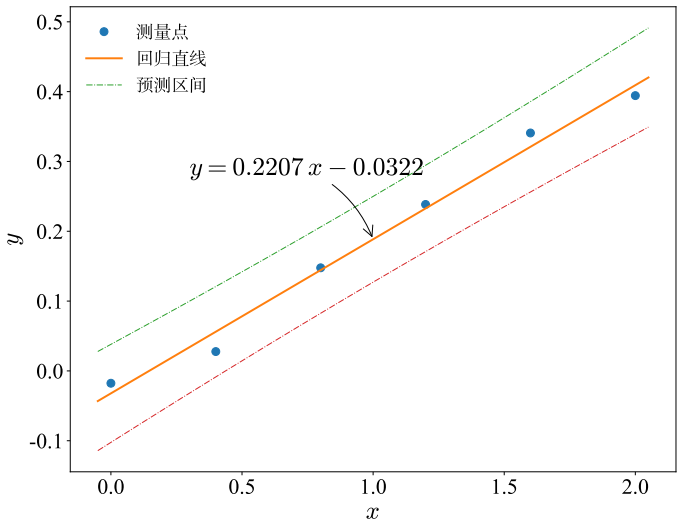


Figure 3.13: The regression equation, the regression line, and the prediction interval.

3.7.4 The Prediction Interval

The prediction interval is the range of values that y may take once a particular value x_0 is given for x. Under the theoretical model, the "true value" of y at x_0 is:

$$y_0 = \beta_1 x_0 + \beta_0 \tag{3.97}$$

Because random error is present, however, only its estimate can be obtained:

$$\hat{y}_0 = \hat{\beta}_1 x_0 + \hat{\beta}_0 \tag{3.98}$$

The actual value of y_0 therefore fluctuates about the center $\hat{y}_0$ and forms an interval, as follows:

$$(\hat{y}_0 - \delta, \quad \hat{y}_0 + \delta) \tag{3.99}$$

Under the classical assumptions of linear regression, if y_0 and $\hat{y}_0$ are independent, then:

$$(y_0 - \hat{y}_0) \sim N\left(0, \left(1 + \frac{1}{n} + \frac{(x_0 - \bar{x})^2}{\sum_{i=1}^{n}(x_i - \bar{x})^2}\right)\sigma^2\right) \tag{3.100}$$

Replacing σ by the residual standard deviation $\hat{\sigma}$ gives the pivotal quantity:

$$\frac{y_0 - \hat{y}_0}{\hat{\sigma} \times \sqrt{1 + \frac{1}{n} + \frac{(x_0 - \bar{x})^2}{\sum_{i=1}^{n}(x_i - \bar{x})^2}}} \sim t_{n-2} \tag{3.101}$$

The half-width of the $1 - \alpha$ prediction interval is then:

$$\delta = \hat{\sigma} \times t_{0.05,n-2} \times \sqrt{1 + \frac{1}{n} + \frac{(x_0 - \bar{x})^2}{\sum_{i=1}^{n}(x_i - \bar{x})^2}} \tag{3.102}$$

The dash-dot lines in Figure 3.13 mark out the prediction interval for y; in shape it flares out at both ends and is narrowest at $\bar{x}$. Equation 3.102 shows that the width of the prediction interval depends on the sample size n, on the sum of squared deviations of x, and on the distance $|x_0 - \bar{x}|$ from x_0 to $\bar{x}$. The farther x_0 lies from $\bar{x}$, the poorer the precision of the prediction. If x_0 falls outside the range of the x_i, the prediction is an extrapolation, and its precision may become very poor; extrapolation should therefore be used with caution in practice.

3.7.5 Estimating x from y

A second use of the regression equation is to estimate x from y. In quantitative chemical analysis this use is sometimes the only purpose for which the equation is established. In instrumental analysis, for instance, a series of standard solutions must be used to establish a linear equation between the concentration of the analyte and the measured response, and the equation is then used to calculate the concentration of an unknown sample from its measured response. Suppose an unknown sample is measured m times and the mean value of y is $\bar{y}_x$. According to Equation 3.98,

$$x = \frac{\bar{y}_x - \hat{\beta}_0}{\hat{\beta}_1} \tag{3.103}$$

The standard error of the estimated value of x is calculated as:

$$s_x = \frac{\hat{\sigma}}{\hat{\beta}_1}\left(\frac{1}{m} + \frac{1}{n} + \frac{(\bar{y}_x - \bar{y})^2}{(\hat{\beta}_1)^2 \sum_{i=1}^{n}(x_i - \bar{x})^2}\right)^{1/2} \tag{3.104}$$

The true-value interval for x is therefore:

$$\mu_x = x \pm t s_x \tag{3.105}$$

Here the t value has $n-2$ degrees of freedom, and its significance level is chosen according to the particular requirement.

Example 3.13 The iron content of an aqueous solution is determined by the 1,10-phenanthroline colorimetric method. Measurement of a series of standard solutions gives the following data:

Variable	Value
Concentration (x), mg/L	0.0000, 0.4000, 0.8000, 1.200, 1.600, 2.000
Absorbance (y)	-0.0177, 0.0277, 0.1477, 0.2384, 0.3408, 0.3943

An unknown sample was also measured 3 times, giving a mean absorbance of 0.290. Establish the regression equation from the experimental data and calculate the concentration of the unknown sample.

Solution: Because no conversion of units is involved, the calculation below carries no units; units are given only in the final result. The relevant quantities are calculated first.

$$\begin{aligned} l_x &= \sum_{i=1}^{6} x_i \\ &= 0.0000 + 0.4000 + 0.8000 + 1.200 + 1.600 + 2.000 \\ &= 6.000 \end{aligned}$$

$$
\begin{aligned}
l_y &= \sum_{i=1}^{6} y_i \\
&= -0.0177 + 0.0277 + 0.1477 + 0.2384 + 0.3408 + 0.3943 \\
&= 1.1312
\end{aligned}
$$

$$
\begin{aligned}
l_{xx} &= \sum_{i=1}^{6} (x_i)^2 \\
&= 0.0000^2 + 0.4000^2 + 0.8000^2 + 1.200^2 + 1.600^2 + 2.000^2 \\
&= 8.800
\end{aligned}
$$

$$
\begin{aligned}
l_{xy} &= \sum_{i=1}^{6} x_i y_i \\
&= 0.000 \times (-0.0177) + 0.4000 \times 0.0277 + \cdots + 2.000 \times 0.3943 \\
&= 0.000 + 0.0111 + 0.1181 + 0.2861 + 0.5452 + 0.7886 \\
&= 1.7491
\end{aligned}
$$

$$
\bar{x} = \frac{l_x}{n} = \frac{6.000}{6} = 1.000, \quad \bar{y} = \frac{l_y}{n} = \frac{1.1312}{6} = 0.1885
$$

The regression coefficients are calculated next:

$$
\begin{aligned}
\hat{\beta}_1 &= \frac{\sum_{i=1}^{6} y_i x_i - \frac{\left(\sum_{i=1}^{6} x_i\right)\left(\sum_{i=1}^{n} y_i\right)}{n}}{\sum_{i=1}^{n} x_i^2 - \frac{\left(\sum_{i=1}^{n} x_i\right)^2}{n}} \\
&= \frac{1.7491 - \frac{6.000 \times 1.1312}{6}}{8.800 - \frac{6.000 \times 6.000}{6}} \\
&= 0.2207
\end{aligned}
$$

$$
\begin{aligned}
\hat{\beta}_0 &= \frac{\sum_{i=1}^{n} y_i}{n} - \hat{\beta}_1 \frac{\sum_{i=1}^{n} x_i}{n} \\
&= \frac{1.1312}{6} - 0.2207 \times \frac{6.000}{6}
\end{aligned}
$$

$$= -0.0322$$

The regression equation is therefore:

$$\hat{y} = 0.2207x - 0.0322$$

The values of $\hat{y}$ calculated from the regression equation are:

$$-0.0322, \quad 0.0561, \quad 0.1444, \quad 0.2327, \quad 0.3209, \quad 0.4092$$

The relevant statistics are computed:

$$\begin{aligned} S_R &= \sum_{i=1}^{n} (\hat{y}_i - \bar{y})^2 \\ &= (-0.0322 - 0.1885)^2 + (0.0561 - 0.1885)^2 + \cdots + (0.4092 - 0.1885)^2 \\ &= 0.1364 \end{aligned}$$

$$\begin{aligned} S_e &= \sum_{i=1}^{n} (y_i - \hat{y}_i)^2 \\ &= (-0.0177 + 0.0322)^2 + (0.0277 - 0.0561)^2 + \cdots + (0.3943 - 0.4092)^2 \\ &= 1.68 \times 10^{-3} \end{aligned}$$

$$F = \frac{S_R}{S_e/(n-2)} = \frac{0.1364}{1.68 \times 10^{-3}/(6-2)} = 3.25 \times 10^2$$

The tabulated value is $F_{0.05,1,4} = 7.71$, so $F \gg F_{0.05,1,4}$, and the regression equation is significant.

From the regression equation it can be calculated that when $\bar{y}_x = 0.290$, the value of x is:

$$x = \frac{\bar{y}_x - \hat{\beta}_0}{\hat{\beta}_1} = \frac{0.290 + 0.0322}{0.2207} = 1.46$$

Since:

$$\hat{\sigma} = \sqrt{\frac{S_e}{n-2}} = \sqrt{\frac{1.68 \times 10^{-3}}{6-2}} = 2.05 \times 10^{-2}$$

$$\sum_{i=1}^{6} (x_i - \bar{x})^2 = (0.0000 - 1.000)^2 + \cdots + (2.000 - 1.000)^2 = 2.800$$

therefore:

$$
\begin{aligned}
s_x &= \frac{\hat{\sigma}}{\hat{\beta}_1}\left(\frac{1}{m}+\frac{1}{n}+\frac{(\bar{y}_x-\bar{y})^2}{(\hat{\beta}_1)^2\sum_{i=1}^{n}(x_i-\bar{x})^2}\right)^{1/2} \\
&= \frac{2.05\times10^{-2}}{0.2207}\left(\frac{1}{3}+\frac{1}{6}+\frac{(0.290-0.1885)^2}{0.2207^2\times2.800}\right)^{1/2} \\
&= 0.0705
\end{aligned}
$$

The tabulated value is $t_{0.05,4} = 2.78$, so the confidence interval for the population mean of x is:

$$
\begin{aligned}
\mu_x &= x \pm t \times s_x \\
&= 1.46 \pm 2.78 \times 0.0705 \\
&= 1.46 \pm 0.20\,(\text{mg/L})
\end{aligned}
$$

Hand calculation of a regression involves a large number of arithmetic operations and is prone to error. Octave provides the regress function, and the appendix of this book also gives the program linearfit.m (which does not include the part that calculates x back from y).

Summary of Key Points

1. **Classification of error.** Systematic error, which can be corrected and which governs accuracy, and random error, which cannot be eliminated but can be quantified and which governs precision, are the prerequisite for understanding all data processing. Total error = systematic error + random error; the relative error relates the deviation to the size of the quantity measured and is the standard index for comparing accuracy across magnitudes.

2. **The normal distribution** $N(\mu, \sigma^2)$**.** The general statistical model for measured data. μ fixes the location (the true value) and σ fixes the spread (the precision). The standardized transformation $U = (X - \mu)/\sigma$ converts any normal distribution into $N(0, 1)$ and is the basis for obtaining probabilities from a table. The interval $\mu \pm 2\sigma$ contains 95.5% of the data, which is the statistical ground for "estimating a true value at the 95% confidence level."
3. **Inferring the population from a sample.** $\bar{x}$ is an unbiased estimate of μ, and $s^2 = \sum(x_i - \bar{x})^2/(n-1)$ is an unbiased estimate of σ^2 (the denominator $n - 1$ being the degrees of freedom). The three sampling distributions, χ^2 for inferences about a variance, t for testing a mean with a small sample ($n \geq 30$ permits the normal approximation), and F for comparing two variances, form the theoretical cornerstone of classical statistical inference.
4. **Rejection of outliers.** Both the International Organization for Standardization and the national standards of China recommend the Grubbs method. The statistic $G = |x_d - \bar{x}|/s$ is compared with the critical value $G_{P,n}$: if $G > G_{\text{crit.}}$ the point is judged an outlier and rejected.
5. **Standard expression of a measurement result.** The standard form is $\bar{x} \pm U$, where $U = ku$ is the expanded uncertainty, $u = s/\sqrt{n}$ is the standard uncertainty, and k is the confidence factor. A complete report states U, u, ν (degrees of freedom), P (confidence level), and k at the same time.
6. **Propagation of error.** The general formula is $u_Y^2 = \sum a_i^2 u_i^2$ ($a_i = \partial g/\partial X_i$). Addition and subtraction look to absolute errors, multiplication and division to relative errors, and the error of the final result is decided by the largest of the component terms.
7. **Significant figures and their arithmetic.** "Complete reading" (every digit the instrument displays) and "estimated reading" (the certain digits plus one estimated digit) are the two ways of taking a reading. Rounding follows the rule of "round half to even." In addition and subtraction the term with the largest absolute error fixes the last digit retained; in mul-

tiplication and division the term with the largest relative error fixes the number of significant figures.

8. **Hypothesis testing.** The t test decides whether a significant difference exists between a mean and a true value, or between two means; the F test decides whether two variances are homogeneous. Type I error (α, a false positive) and type II error (β, a false negative) trade off against each other, and $1-\beta$ is the power of the test.
9. **Linear regression.** The method of least squares yields $\hat{\beta}_1$ and $\hat{\beta}_0$ that minimize the sum of squares of residuals. An F test and the coefficient of determination R^2 are used to judge the significance of the regression equation. The regression equation is the basis of a calibration method, converting an instrument signal into an analytical result.

Self-Check

Self-Check: after reading this chapter you should be able to:

Basic level (Can you do it?)

- distinguish systematic from random error in the fluctuation of a balance reading, and give two examples of each from different sources
- given μ and σ, use a table of the standard normal distribution to calculate the probability that a measured value lies within $\mu \pm k\sigma$ ($k = 1, 2, 3$)
- apply the Grubbs method to test a set of measured data for an outlier, reading the critical value from a table and reaching the right judgment
- express the result of a set of data correctly: give $\bar{x} \pm U$ and note u, ν, P, and k alongside it
- round numbers by the rule of "round half to even," and complete addition and subtraction as well as multiplication and division under the rules for significant figures
- calculate the standard deviation of an indirectly measured quantity, such as

a standardized concentration, applying the error propagation formula correctly

- use the t test to judge whether a significant difference exists between a sample mean and a given true value
- given two sets of corresponding x and y data, establish a linear regression equation and complete the F test of significance

Deeper level (Can you explain it?)

- Why is the denominator of s^2 equal to $n-1$ rather than n? Explain it to a classmate from the viewpoint that "one degree of freedom has been lost."
- How do the curve shapes of the t distribution and the normal distribution differ, and why must the t distribution be used for a small sample instead of the normal distribution?
- What is the relation between a "type I error" and a "type II error"? Why does analytical chemistry generally pay more attention to avoiding a type I error?

Exercises

3.1 Explain the concepts of the population and the sample.

3.2 What type of error does each of the following situations cause? If the error is systematic, how should it be eliminated?

(1) Corrosion of the weights.

(2) The sample absorbs moisture from the air during weighing.

(3) The zero point of the balance shifts slightly.

(4) The last digit is estimated when a burette reading is taken.

(5) The reagent contains a trace of the analyte.

(6) Impurities coprecipitate in a gravimetric determination.

3.3 How many significant figures does each of the following data have?

(1) 0.068 (2) 21.080 (3) 6.4×10^{-3}

(4) 6000 (5) 99.50 (6) 101.0

3.4 Are the following reports correct? Why?

(1) A 0.50 g sample containing sodium chloride is weighed, and analysis gives a sodium chloride content of 36.68%. (2) 29.418 g of $K_2Cr_2O_7$ is weighed and made up to a 1 L solution in a volumetric flask, and its concentration is expressed as 0.1 mol/L.

★★ 3.5 A primary standard substance, $KHC_8H_4O_4$ (potassium hydrogen phthalate), is weighed on an analytical balance: 1.0216 g (readability 0.0001 g). It is dissolved in water and diluted to the mark in a 100.0 mL volumetric flask. A 25.00 mL portion of this solution is taken and titrated with NaOH solution to the end point with phenolphthalein as the indicator, and 24.52 mL of the NaOH solution is consumed. Calculate the concentration of this NaOH solution (in mol/L), and state how many significant figures the result should retain.

(Given $M(KHC_8H_4O_4) = 204.22$ g/mol.)

★★ 3.6 Carry out the following calculations under the rules for significant figures:

(1) $12.345 + 1.6574 - 0.0954 + 1.57$

(2) $1.781 \times 0.458 + 6.9 \times 10^{-5} - 0.0623 \times 0.00418$

(3) $\dfrac{7.728 \times 62.50}{0.006451 \times 216.3}$

(4) $\sqrt{\dfrac{1.5 \times 10^{-8} \times 6.1 \times 10^{-8}}{3.3 \times 10^{-6}}}$

(5) $\dfrac{1.45 \times 10^{-5} \times 5.23 \times 10^{-9}}{2.3 \times 10^{-5}}$

(6) $\dfrac{\pi \times 35.7^3}{3.45 \times 10^{-5}}$

(7) the hydrogen ion concentration for pH = 0.25.

3.7 The percentage of iron in a sample was determined 6 times, with the results 33.11%, 33.45%, 33.38%, 33.24%, 33.28%, and 33.29%. Calculate: (1) the mean of the analytical results; (2) the standard deviation and the relative standard deviation.

3.8 A balance has a weighing error of ±0.1 mg. If the error of a weighing is required to be 0.1%, what is the smallest mass that should be weighed?

★★ 3.9 A pharmaceutical plant produces calcium tablets for adults and for young children. A new analytical method is now applied to determine both kinds of tablet, and two sets of data are obtained:

Type	Content (mg)
Adult	947.65, 953.79, 952.69, 950.35, 954.02, 944.51, 949.45, 946.62, 958.14
Young children	452.31, 453.86, 447.04, 448.69, 449.24, 453.08, 449.22, 451.96, 444.25

Calculate the mean, the standard deviation, and the relative standard deviation of these two sets of data. Compare the standard deviation and the relative standard deviation of the method as applied to the two kinds of tablet.

★★ 3.10 Given $R = 23.6(\pm 0.2) + 0.184(\pm 0.006) = 23.784$, where the value in parentheses is the standard deviation, find for R: (a) the standard deviation; (b) the relative standard deviation; (c) the expression after rounding to the proper number of digits.

★★ 3.11 The mass fraction of iron in a sample was determined by complexometric titration, and the mean of six results was 46.20%; the same sample was determined by a redox method, and the mean of four results was 46.02%. If the standard deviation of a single determination is 0.08% for both methods, decide whether the results obtained by the two methods differ significantly at the 95% confidence level. Given $t_{0.05,8} = 2.31$, $t_{0.05,9} = 2.26$, $t_{0.05,10} = 2.23$.

★★★ 3.12 The blood glucose concentration (mmol/L) of a patient was determined

by electroanalysis, and 10 results were 7.5, 7.4, 7.7, 7.6, 7.5, 7.6, 7.6, 7.5, 7.6, and 7.6. Find the relative standard deviation of this set of data and the confidence interval at the 95% confidence level, and decide whether this result differs significantly from the blood glucose concentration of a healthy person, 6.7 mmol/L. Given a confidence factor $k = 1.96$.

★★★ 3.13 The mass fraction of lead w(Pb) in a sample was determined by atomic absorption spectrometry and by voltammetry, with the following results:

	Atomic absorption method	Voltammetric method
Mean	$\bar{x}_1 = 0.74\%$	$\bar{x}_2 = 0.70\%$
Standard deviation	$s_1 = 0.05\%$	$s_2 = 0.06\%$
Determinations	$n_1 = 6$	$n_2 = 6$

Decide whether the results obtained by the two methods differ significantly at the significance level $\alpha = 0.10$. Given the one-tailed $F_{0.05,5,5} = 5.05$ and the two-tailed $t_{0.10,10} = 1.81$.

★★★ 3.14 The mass fraction of Fe_2O_3 in an ore was determined four times by a redox method and four times by a photometric method, the results being $\bar{x}_1 = 0.60\%$, $s_1 = 0.07\%$ and $\bar{x}_2 = 0.68\%$, $s_2 = 0.03\%$, respectively. Decide whether a systematic error exists between the two methods at the significance level 0.10. Given the one-tailed $F_{0.05,3,3} = 9.28$ and the two-tailed $t_{0.10,6} = 1.94$.

★★ 3.15 For an industrial soda ash the product is acceptable when the total alkalinity, expressed as Na_2CO_3, satisfies $w(Na_2CO_3) \geq 80.0\%$, and the standard deviation of the analytical method is $\sigma = 0.09\%$. After the production process was modified, the mean of five determinations was 80.15%. Has the total alkalinity increased markedly at the significance level 0.05? Given $u_{0.05} = 1.96$ and $u_{0.10} = 1.64$.

★★ 3.16 The population standard deviation of an enzymatic determination of blood glucose concentration is $\sigma = 0.11$ mmol/L. If the confidence interval of the mean is required to be $\bar{x} \pm 0.1$ mmol/L at the 95% confidence level ($u = 1.96$), how many replicate determinations are needed at least?

★★★ 3.17 The DDT content (mg/g) in the leaves of a plant that had not been sprayed with DDT was determined by a certain method, giving 0.2, 0.4, 0.8, 0.5, and 0.2. A second leaf sample gave DDT contents (mg/g) of 0.4, 0.5, 0.8, 1.0, and 0.5. Decide whether this plant had been sprayed with DDT. The values $t_{0.05,8} = 2.31$ and $t_{0.10,8} = 1.86$ are known.

★★★ 3.18 In a training exercise on the analytical balance, a student transferred a solid sample from a weighing bottle into a weighed crucible and compared the loss in mass of the weighing bottle with the gain in mass of the crucible. One student performed the operation 6 times, with the following results (in g):

	Data
Mass loss of the weighing bottle	0.2125, 0.3026, 0.1426, 0.1573, 0.1443, 0.1561
Mass gain of the crucible	0.2124, 0.3025, 0.1423, 0.1568, 0.1443, 0.1566

Decide at the significance level 0.05 whether the student's weighing is acceptable.

★★ 3.19 The standard deviation of a balance weighing is known to be $s_m = 0.1$ mg, and the standard deviation of each reading of a burette is $s_V = 0.02$ mL. A primary standard has a molar mass of 105.00 g/mol. A 0.5000 g portion of this primary standard is weighed out to standardize a certain solution, and the titration to the end point consumes 32.52 mL of the solution. Calculate the standard deviation of the concentration of the resulting solution.

3.20 A reagent solution of 100.0 mL is to be prepared. Which procedure is better: (1) measuring twice with a 50 mL pipet (measuring error 0.05 mL), or (2) measur-

ing four times with a 25 mL pipet (measuring error 0.03 mL)?

★★★ 3.21 Calculate the following expression:

$$\frac{(40.36(\pm 0.02) - 10.23(\pm 0.02)) \times 0.153(\pm 0.001)}{(500(\pm 7) + 250(\pm 5)) \times 40.00(\pm 0.01)}$$

★★ 3.22 A 0.0010 mol/L solution is to be prepared from a 1.0 mol/L stock solution, and two procedures are available: (1) measure the stock solution with a 1 mL pipet into a 1000 mL volumetric flask and dilute; (2) first measure the stock solution with a 20 mL pipet into a 1000 mL volumetric flask, dilute, then measure this dilute solution with a 25 mL pipet into a 500 mL volumetric flask and dilute. The standard deviation of the 1 mL pipet is 0.006 mL, that of the 1000 mL volumetric flask is 0.3 mL, that of both the 20 mL and the 25 mL pipets is 0.03 mL, and that of the 500 mL volumetric flask is 0.2 mL. Calculate the relative standard deviation of the two preparation procedures and state which one introduces the smaller error.

★★★ 3.23 The molar mass of a gas is calculated from

$$M = \frac{gRT}{PV}$$

where g is the mass of the gas (g), R is the molar gas constant, T is the temperature (K), P is the gas pressure (atm), and V is the gas volume (L). The measured data are:

$$g = 0.118(\pm 0.002), \quad R = 0.082056(\pm 0.000001), \quad T = 298.2(\pm 0.1),$$
$$P = 0.724(\pm 0.005), \quad V = 0.250(\pm 0.005)$$

Find: (1) the molar mass of the gas; (2) which quantities should be measured more precisely if the accuracy of the molar mass is to be improved.

★★★ 3.24 The mass fraction of iron in an iron ore was determined by the dichromate method. A 2.2405 g sample was taken and made up to 250.0 mL of solution;

a 25.00 mL aliquot was titrated each time. The titer of the $K_2Cr_2O_7$ standard solution toward iron is $T(Fe/K_2Cr_2O_7) = 5.585\,mg/mL$, and the volumes of the standard solution consumed in five determinations were 24.63, 24.67, 24.64, 24.61, and 24.65 mL. The standard deviation of the weighing is known to be 0.1 mg, and that of the titer is 6×10^{-6} g/mL. Calculate the mass fraction of iron $w(Fe)$ and its relative standard deviation.

★★★ 3.25 A 25.00 mL (V_1) NaOH solution was standardized with 0.01250 mol/L (c_2) HCl standard solution, and 20.00 mL (V_2) of the HCl solution was consumed. The standard deviation of a pipet delivery is known to be $s_1 = 0.02$ mL, and that of each reading of the burette is $s_2 = 0.01$ mL. Calculate the standard deviation of the NaOH concentration c_1 and express the standardization result properly.

★★★ 3.26 An HCl solution was standardized with Na_2CO_3. A mass of Na_2CO_3 of (0.9674 ± 0.0009) g was weighed out, and the titration consumed 27.35 ± 0.04 mL of the HCl solution. Given that $M(Na_2CO_3) = 105.988 \pm 0.001$ g/mol, express the concentration of the standardized HCl solution properly.

★★ 3.27 In titrimetric analysis the mass fraction of component A is calculated from

$$w_A = \frac{cVM_A}{m}$$

where c and V are the concentration of the standard solution and the volume consumed, m is the mass of the sample, and M_A is the molar mass of component A. The standard deviations of c, V, and m are known to be 0.0002 mol/L, 0.02 mL, and 0.0002 g. When $c = 0.05000$ mol/L, $V = 25.00$ mL, $m = 0.2000$ g, and $M_A = 106.0$ g/mol, calculate the standard deviation of w_A and state which factor governs it.

★★ 3.28 The protein content of a sample was determined six times in parallel,

with the results 8.32%, 8.44%, 8.38%, 8.86%, 8.41%, and 8.35%. At the confidence level $P = 95\%$ ($G_{0.95,6} = 1.887$), decide whether an outlier is present in this data set. If so, reject it and recalculate the mean and the standard deviation.

★★★ 3.29 The population mean of the copper content of a certified reference material is known to be $\mu = 25.00\%$ and the population standard deviation $\sigma = 0.12\%$. Assuming that the measured values follow a normal distribution, calculate: (1) the probability that a single measured value falls in the range $24.76\% \sim 25.24\%$; (2) the probability that a single measured value exceeds 25.36%.

★★ 3.30 Two analysts determined the chlorine content of the same sample by the same method. The first made 6 determinations with a standard deviation $s_1 = 0.052\%$, and the second made 5 determinations with a standard deviation $s_2 = 0.031\%$. Does the measurement precision of the two analysts differ significantly? ($\alpha = 0.05$, $F_{0.05,5,4} = 6.26$)

★★★ 3.31 The concentration of a dye was determined spectrophotometrically. A standard series was prepared and the absorbances were measured, with the following data:

Concentration c (mg/L)	0.00	2.00	4.00	6.00	8.00
Absorbance A	0.002	0.162	0.318	0.484	0.640

(1) Establish the linear regression equation; (2) calculate the correlation coefficient r; (3) an unknown sample gives an absorbance $A = 0.400$; find its concentration.

Answer

3.1 (Omitted)

3.2 (Omitted)

3.3 2, 5, 2, 1, 4, 4

3.4 Neither is correct.

3.5 (Omitted)

3.6 (1) 15.48, (2) 0.816, (3) 346.1, (4) 1.7×10^{-5}, (5) 3.3×10^{-9}, (6) 4.14×10^{9}, (7) 0.56 mol/L.

3.7 (1) 33.29%; (2) 0.12% (standard deviation) and 0.36% (relative standard deviation).

3.8 100 mg

3.9 Adult tablets 950.80 mg, 4.27 mg, 0.45%; tablets for young children 449.96 mg, 3.13 mg, 0.70%

3.10 0.2, 0.8%, 23.8($\pm$0.2)

3.11 $t = 3.49 > t_{0.05,8} = 2.31$; a significant difference exists between the two means.

3.12 Relative standard deviation 1.1%, confidence interval (7.56 ± 0.05) mmol/L; $u = 32.2 > 1.96$, so the patient's blood glucose may be judged abnormal.

3.13 $F = 1.44 < 5.05$, $t = 1.25 < 1.81$; the results obtained by the two methods do not differ significantly.

3.14 $F = 5.44 < 9.28$, $t = 2.10 > 1.94$; a systematic error exists between the two methods.

3.15 $u = 3.73 > 1.64$; the total alkalinity has increased markedly.

3.16 At least five replicate determinations are needed.

3.17 $t = 1.39 < t_{0.10,8} = 1.86$, so the DDT content in the leaves under test cannot be considered higher than that in the unsprayed sample; the plant had not been sprayed with DDT.

3.18 $t = 0.61 < t_{0.05,5} = 2.57$, no systematic deviation is detected, so the student's weighing may be judged acceptable.

3.19 $s_c = 0.0001\,\text{mol/L}$

3.20 The second procedure is adopted.

3.21 The result is: $1.54(\pm 0.02) \times 10^{-4}$.

3.22 The relative standard deviations of the two procedures are 0.6% and 0.2%, respectively, and the second procedure introduces the smaller error.

3.23 (1) $M = 16.0 \pm 0.4\,\text{g/mol}$; (2) the precision of the volume and mass measurements should be improved first.

3.24 $w(\text{Fe}) = 61.42\%$, and the relative standard deviation is 0.11%.

3.25 $s_{c_1} = 1.1 \times 10^{-5}\,\text{mol/L}$, $c(\text{NaOH}) = (0.010000 \pm 0.000011)\,\text{mol/L}$.

3.26 $c(\text{HCl}) = (0.6675 \pm 0.0012)\,\text{mol/L}$.

3.27 $s_{w_A} = 0.0028$; among the relative standard deviations the concentration term is the largest and is the governing factor.

3.28 8.86% is an outlier; after its rejection $\bar{x} = 8.38\%$ and $s = 0.05\%$.

3.29 (1) 95.5%; (2) 0.13%.

3.30 $F = 2.81 < F_{0.05,5,4} = 6.26$, so the difference is not significant.

3.31 (1) $A = 0.0800c + 0.001$; (2) $r \approx 0.9999$; (3) 4.99 mg/L.

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